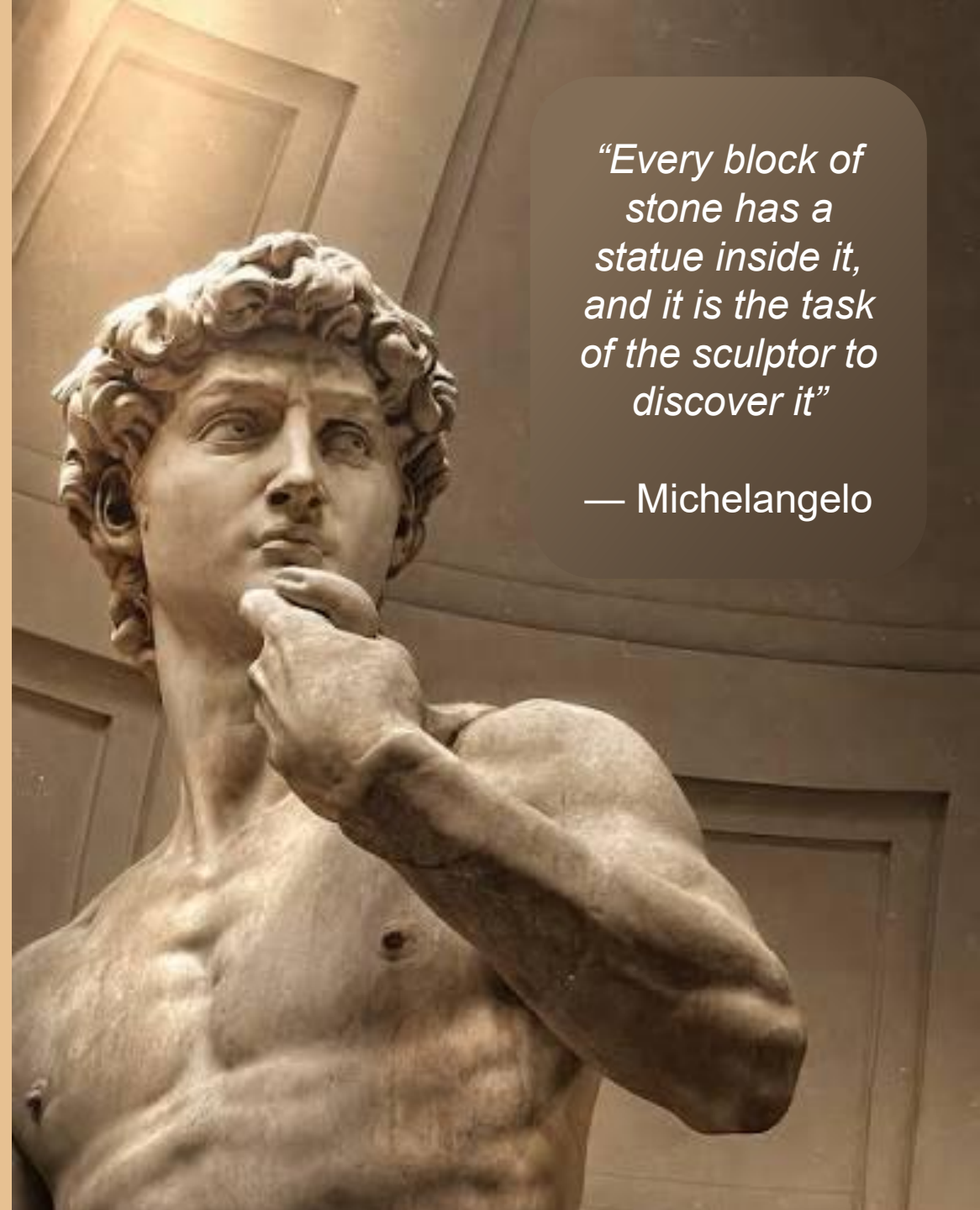


Generative AI / Foundation Models

Dr Varun Ojha

varun.ojha@newcastle.ac.uk
<https://ojhavk.github.io/>

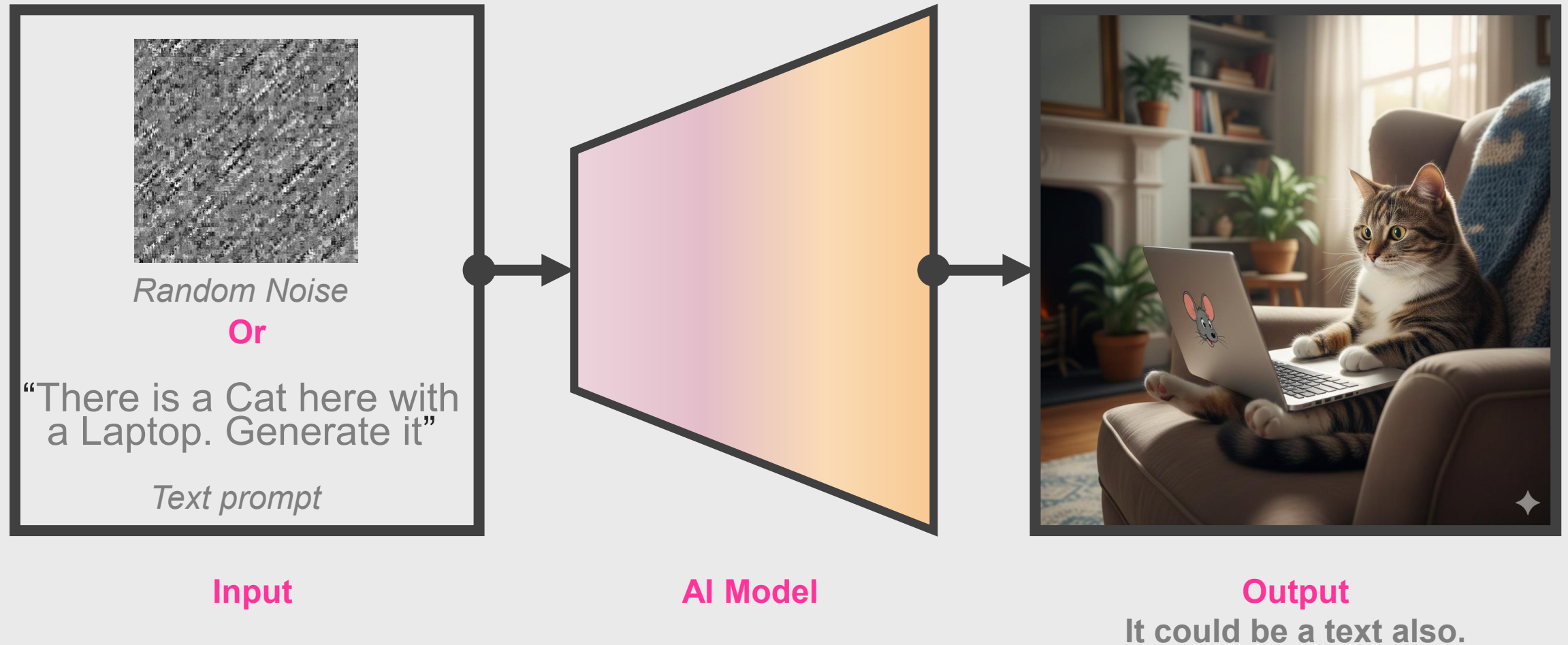
School of Computing
Newcastle University



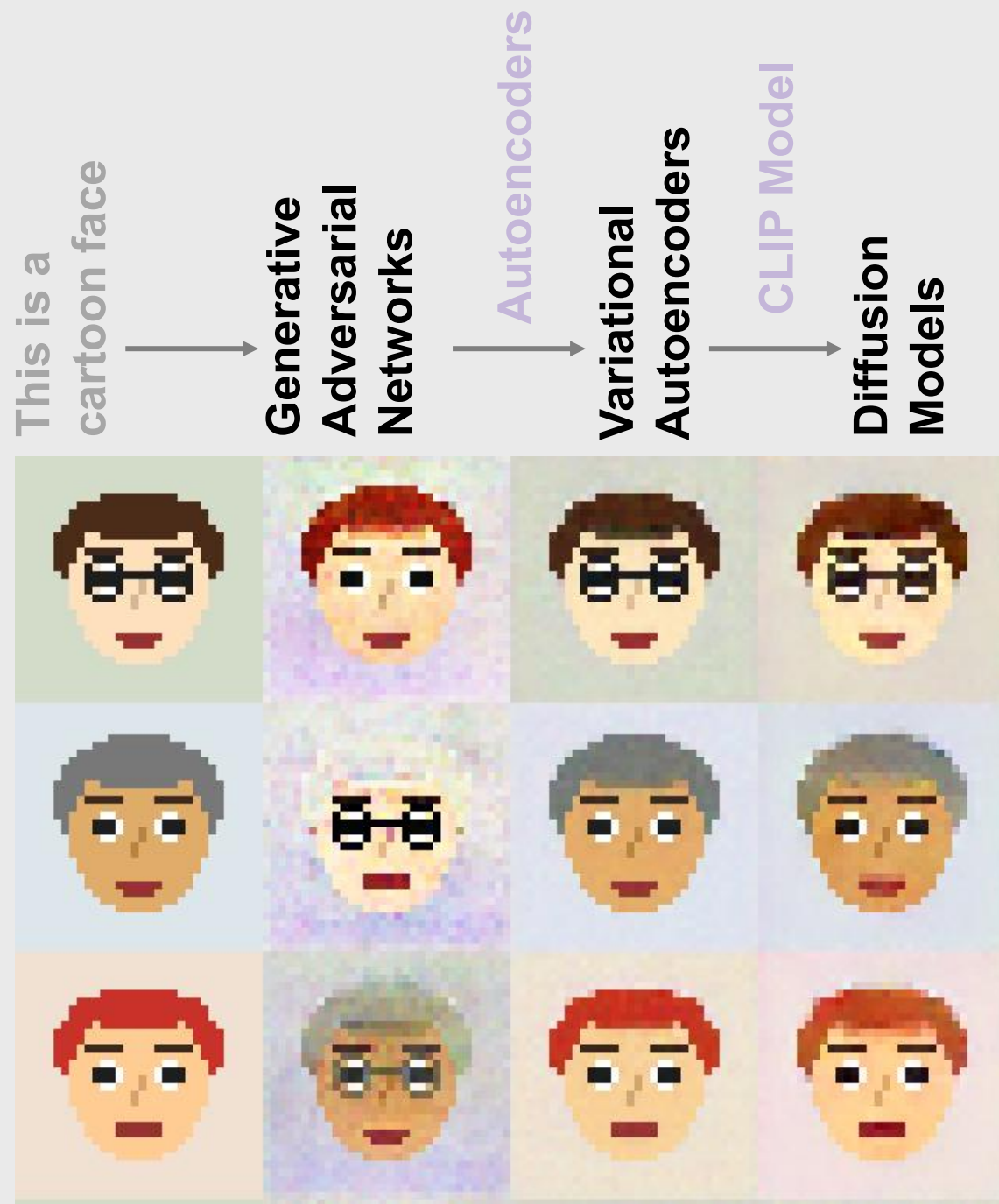
*“Every block of
stone has a
statue inside it,
and it is the task
of the sculptor to
discover it”*

— Michelangelo

Generative AI / Foundation Models



Content



Generative Adversarial Nets

**Ian J. Goodfellow, Jean Pouget-Abadie*, Mehdi Mirza, Bing Xu, David Warde-Farley,
Sherjil Ozair[†], Aaron Courville, Yoshua Bengio[‡]**

Département d'informatique et de recherche opérationnelle
Université de Montréal
Montréal, QC H3C 3J7

Abstract

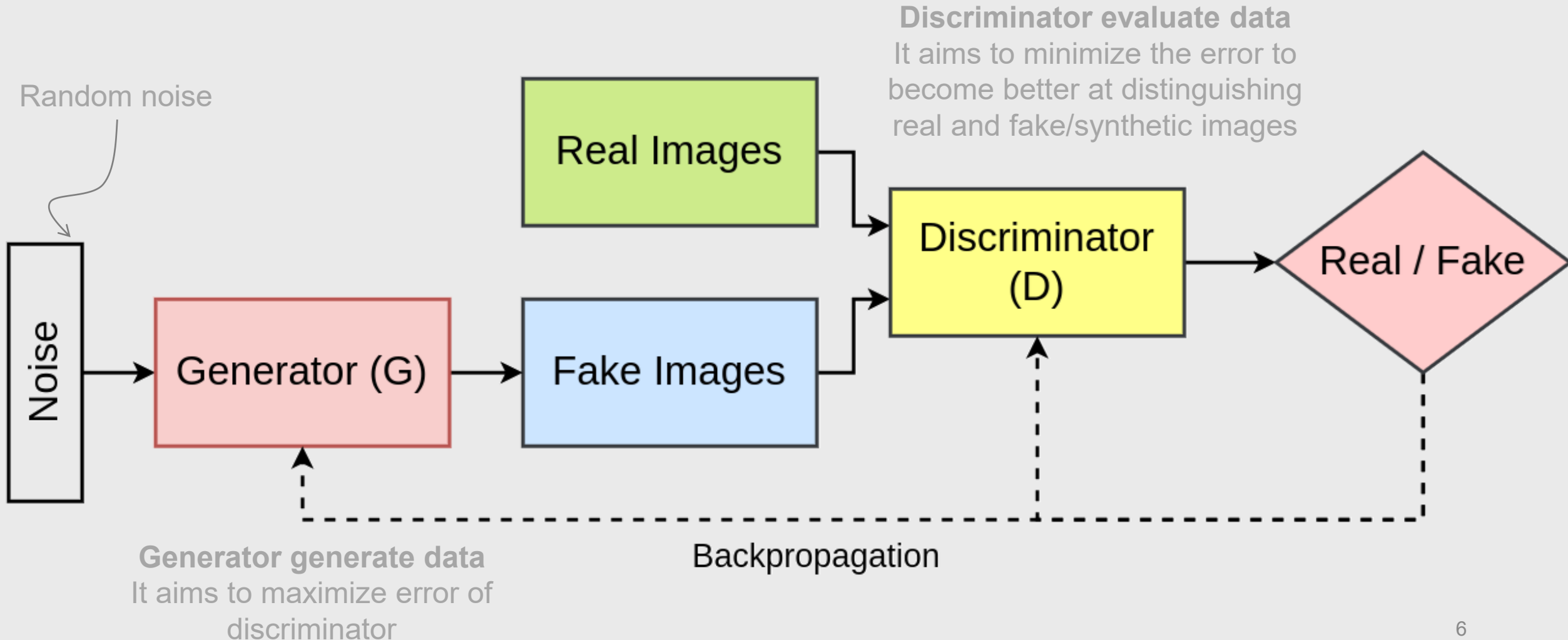
We propose a new framework for estimating generative models via an adversarial process, in which we simultaneously train two models: a generative model G that captures the data distribution, and a discriminative model D that estimates the probability that a sample came from the training data rather than G . The training procedure for G is to maximize the probability of D making a mistake. This framework corresponds to a minimax two-player game. In the space of arbitrary functions G and D , a unique solution exists, with G recovering the training data distribution and D equal to $\frac{1}{2}$ everywhere. In the case where G and D are defined by multilayer perceptrons, the entire system can be trained with backpropagation. There is no need for any Markov chains or unrolled approximate inference networks during either training or generation of samples. Experiments demonstrate the potential of the framework through qualitative and quantitative evaluation of the generated samples.

https://papers.nips.cc/paper_files/paper/2014/hash/f033ed80deb0234979a61f95710dbe25-Abstract.html

Generative Adversarial Networks

9:46 AM

The aim of the GAN generates realistic data from



Generative Adversarial Networks

Image: Bishop, Deep Learning



Random noise

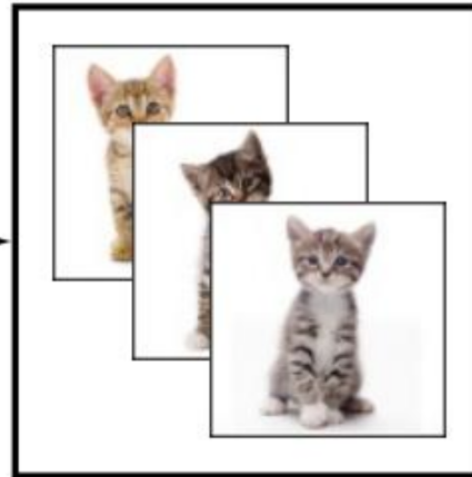
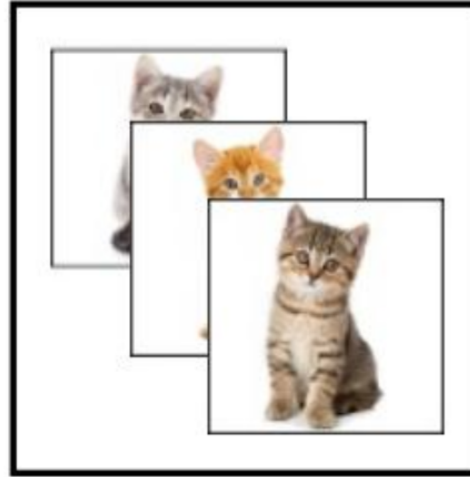
$\mathbf{z}$

Generator

$g(\mathbf{z}, \mathbf{w})$

Generator aims to maximize error of discriminator

real images



synthetic images

Discriminator

$d(\mathbf{x}, \phi)$

Discriminator, is a **binary classifier** that aims to minimize the error to become better at distinguishing real and fake/synthetic images

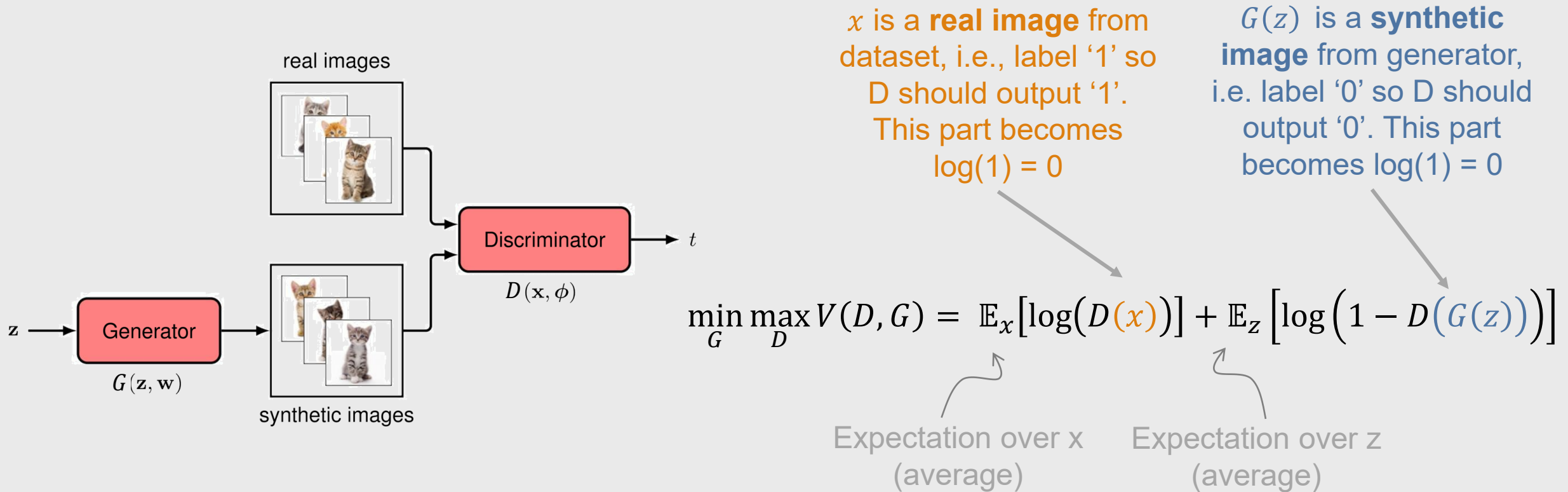
t

9:46 AM

Generative Adversarial Networks

9:46 AM

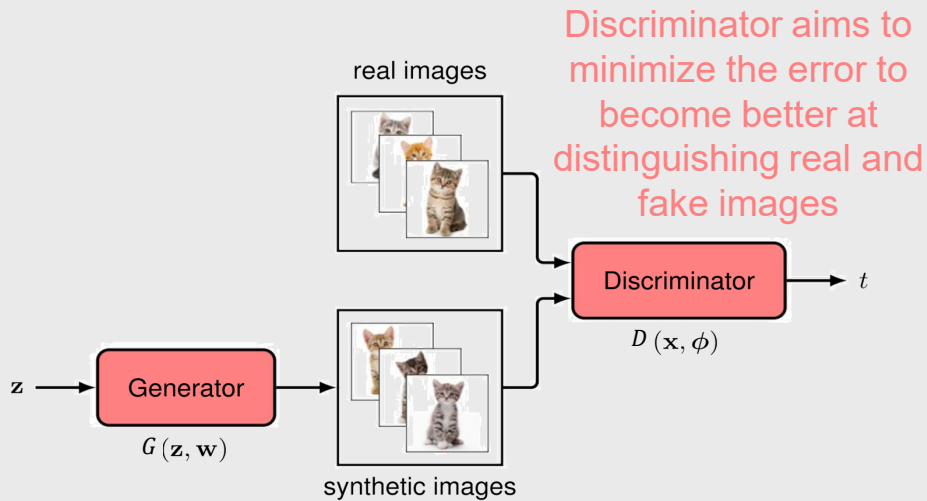
min-max loss



Generative Adversarial Networks

9:46 AM

min-max loss



D is a function of $\log(D(x))$ vs (x)

D is a function of $\log(1 - D(G(z)))$ vs $D(G(z))$

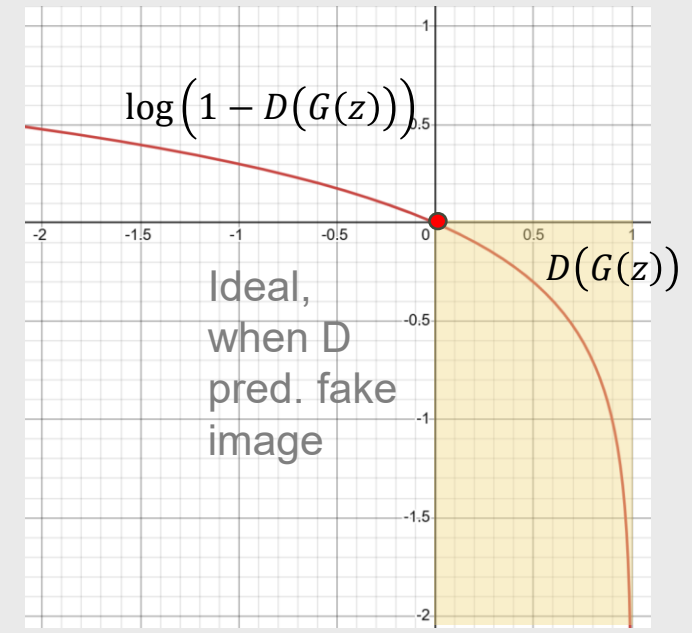
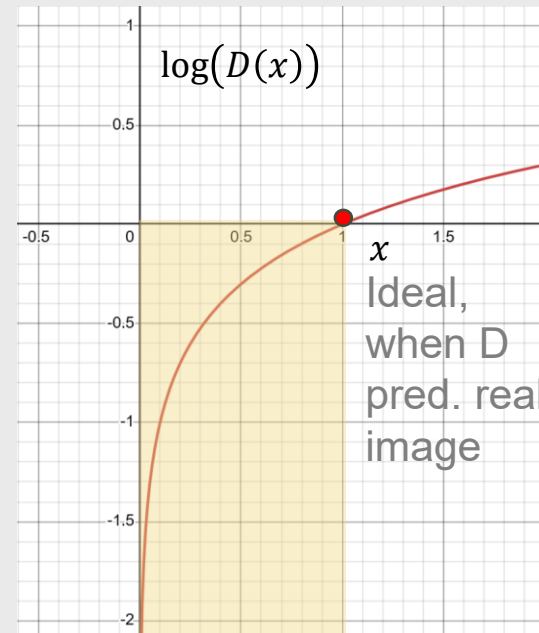
$$\min_G \max_D V(D, G) = \mathbb{E}_x [\log(D(x))] + \mathbb{E}_z [\log(1 - D(G(z)))]$$

D to maximize the probability of assigning the correct label to both training examples and samples from G by maximizing:

$$\mathbb{E}_x [\log(D(x))] + \mathbb{E}_z [\log(1 - D(G(z)))]$$

Low value means discriminator can identify the real vs fake i.e., if $D(x) = 1$ and $D(G(z)) = 0$, then we have $\log 1 + \log 1 = 0 \rightarrow$ **Discriminator Wins.**

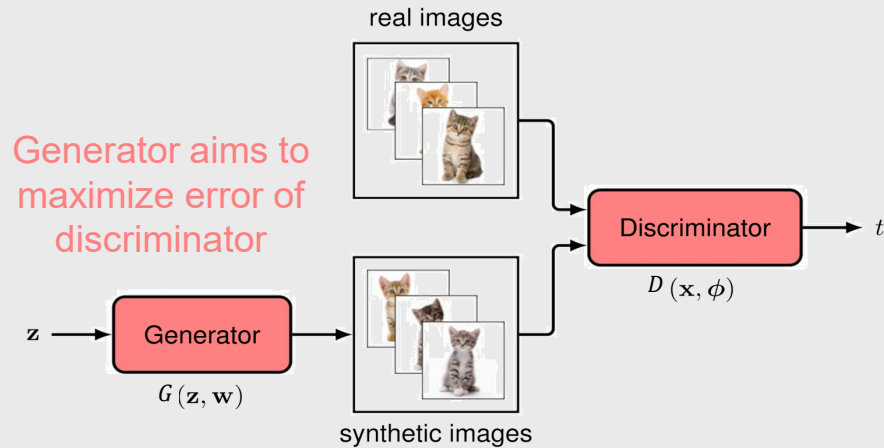
Ideal, when D pred. fake image



Generative Adversarial Networks

9:46 AM

min-max loss



G to minimize:

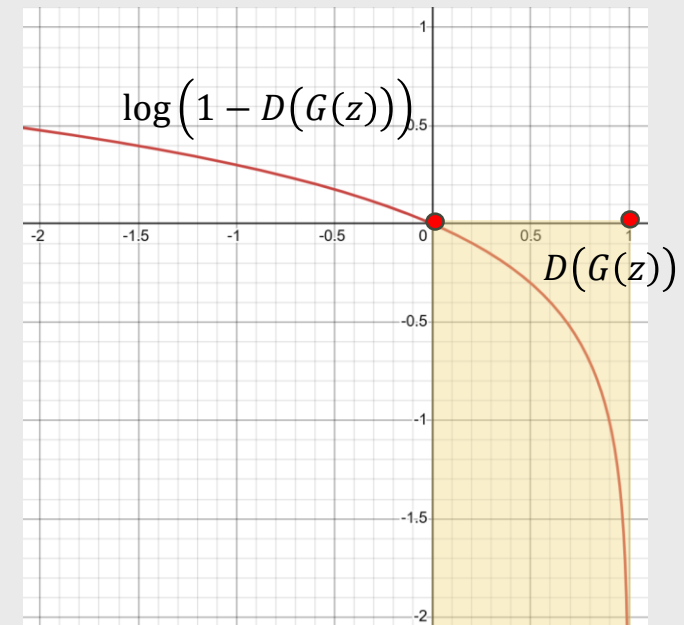
$$\mathbb{E}_z [\log(1 - D(G(z)))]$$

Low value means generator create realistic images, i.e., if $D(G(z))$ close to 1 then generator fools the Discriminator, and Generator Wins

We would $\log(1 - 0.99) = \log(0.01) = -2 \rightarrow$ **Generator Wins**.

G is a function of
 $\log(1 - D(G(z)))$ vs $D(G(z))$

$$\min_G V(D, G) = \mathbb{E}_z [\log(1 - D(G(z)))]$$



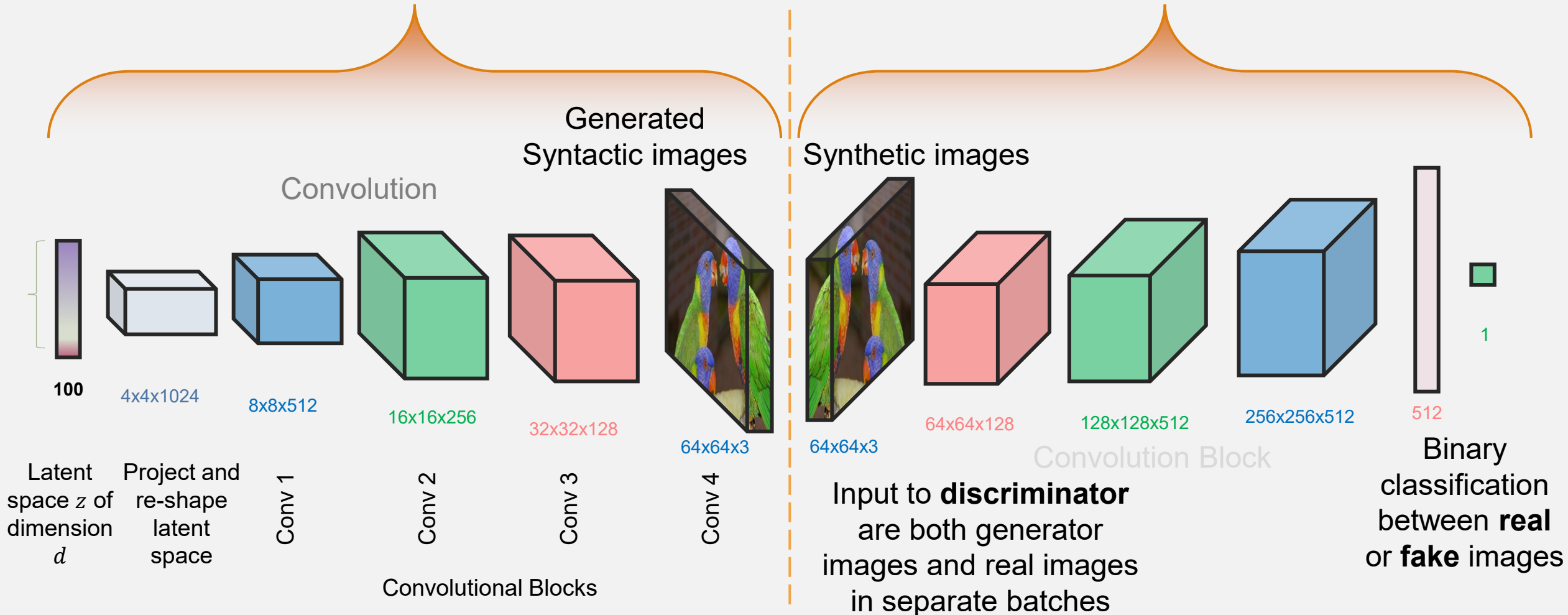
Deep Convolutional GAN (DCGAN)

9:46 AM

The task of the generator is to produce data which the discriminator predicts as being 'real', meaning that it closely resembles the training dataset.

Generator

Discriminator



Super Realistic Generative Adversarial Networks

bicubic
(21.59dB/0.6423)



SRResNet
(23.53dB/0.7832)



SRGAN
(21.15dB/0.6868)



original



StyleGAN to Generate China City Scape

9:46 AM

One of my student's project



(Jia 2022)

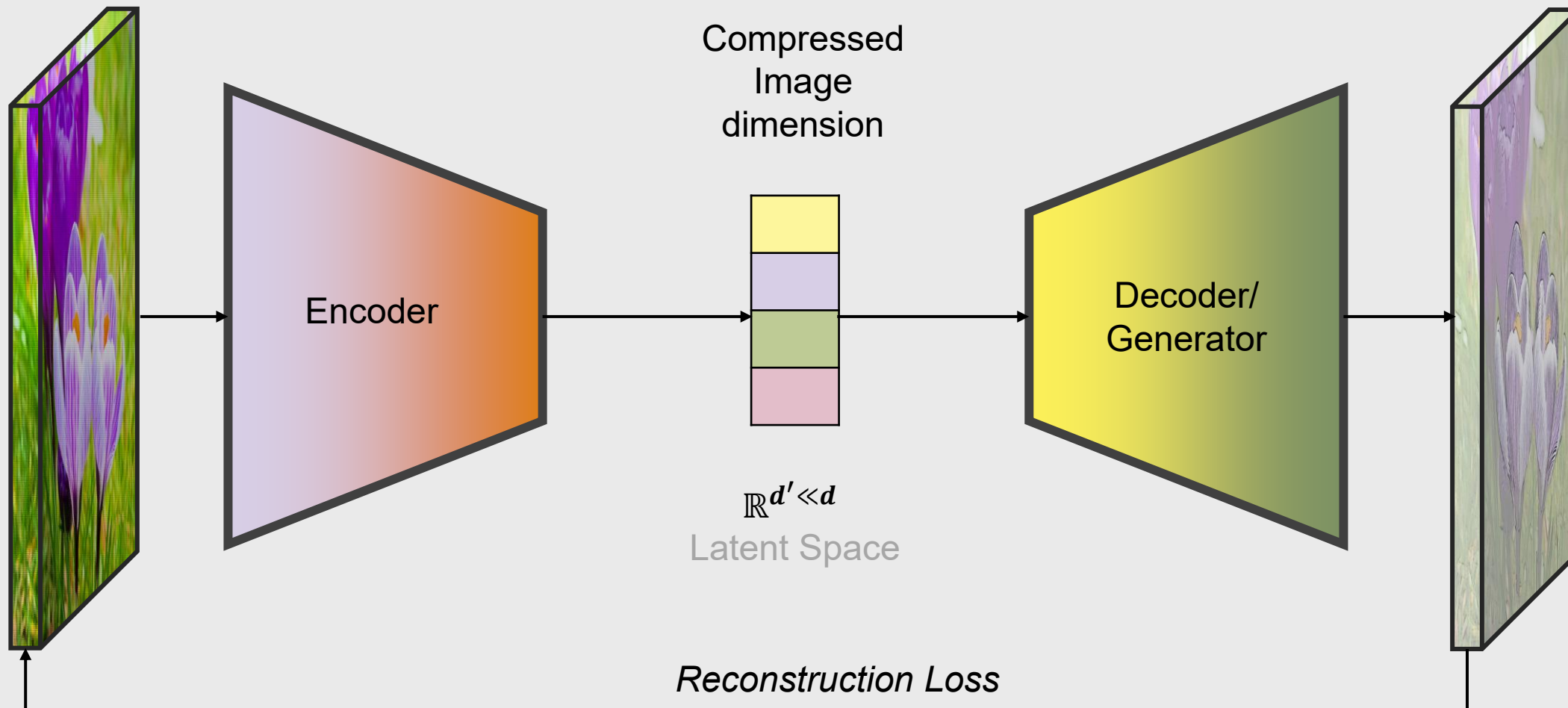
Autoencoder

9:46 AM

Can be used for dimensionality reduction as well.

Input
Image $\mathbf{x} \in \mathbb{R}^d$

Reconstructed
Image $\mathbf{x}' \in \mathbb{R}^{\leq d}$

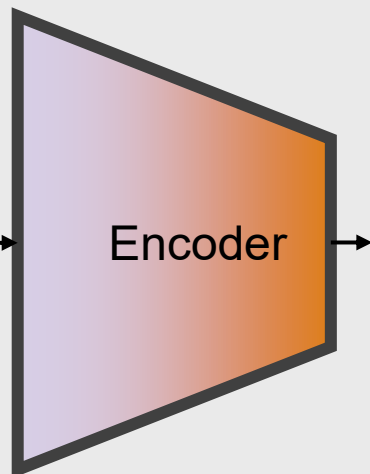
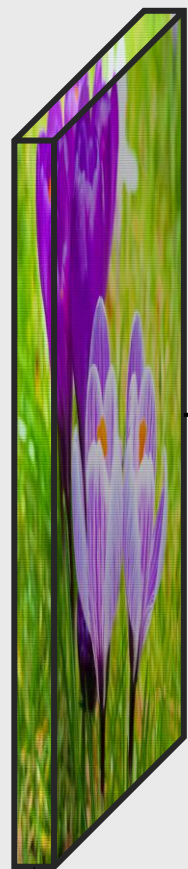


Autoencoder

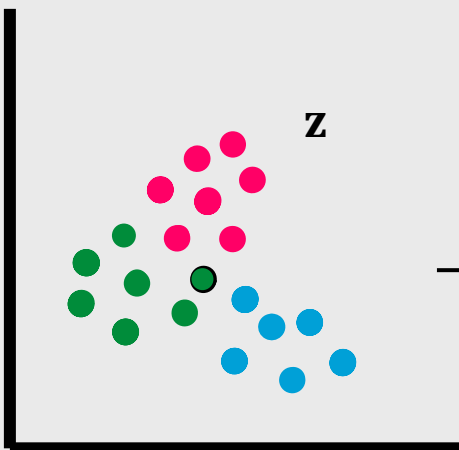
9:46 AM

Can be used for dimensionality reduction as well.

Input
Image $\mathbf{x} \in \mathbb{R}^d$

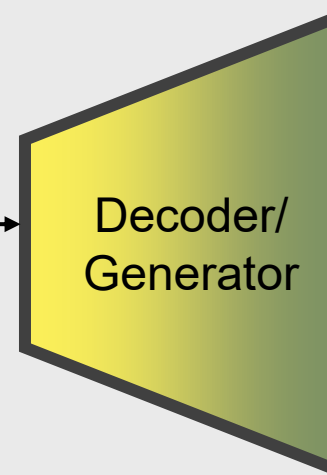


Compressed
Image
dimension

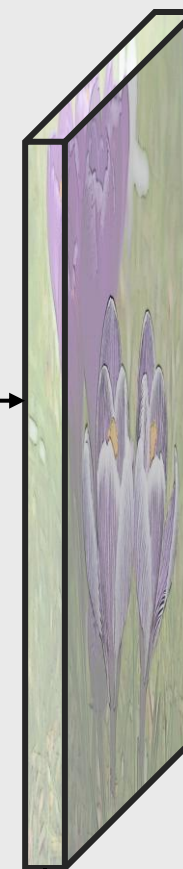


$\mathbb{R}^{d' \ll d}$

Latent Space



Reconstructed
Image $\mathbf{x}' \in \mathbb{R}^{\leq d}$



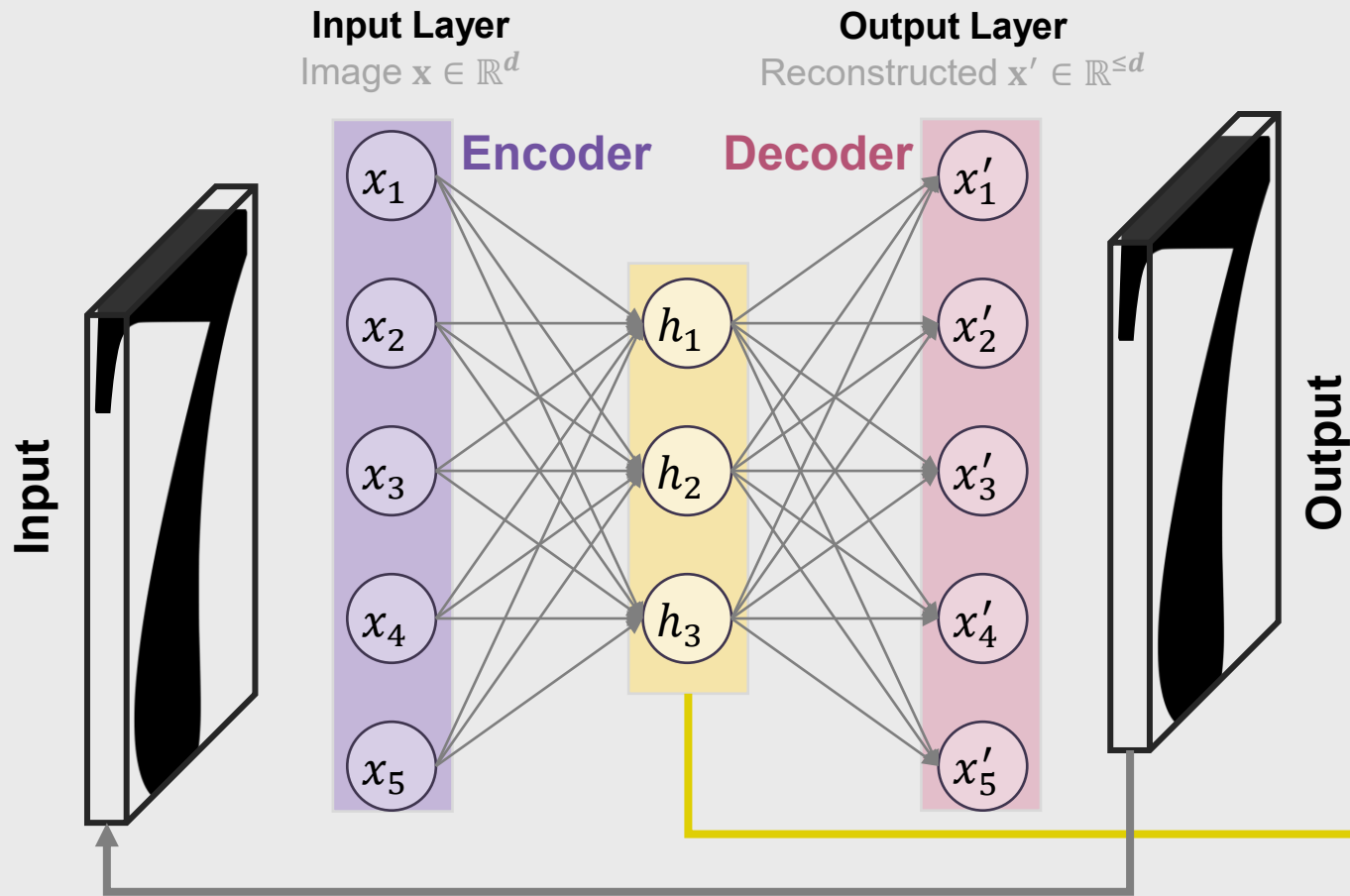
Reconstruction Loss



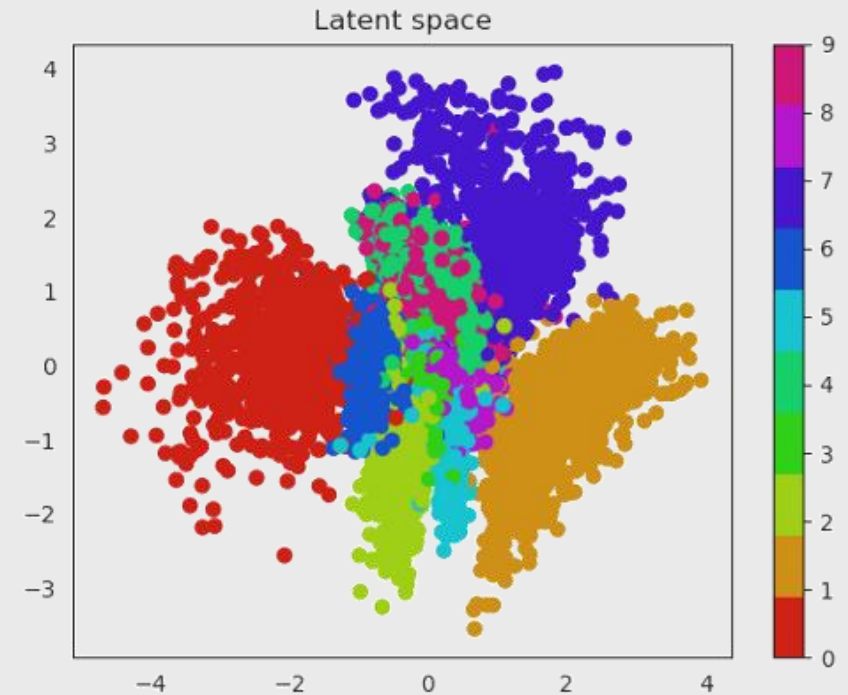
Autoencoder

9:46 AM

Autoencoder takes its feedback/error correction signal from the input itself This it is **Unsupervised Or Self-Supervised**



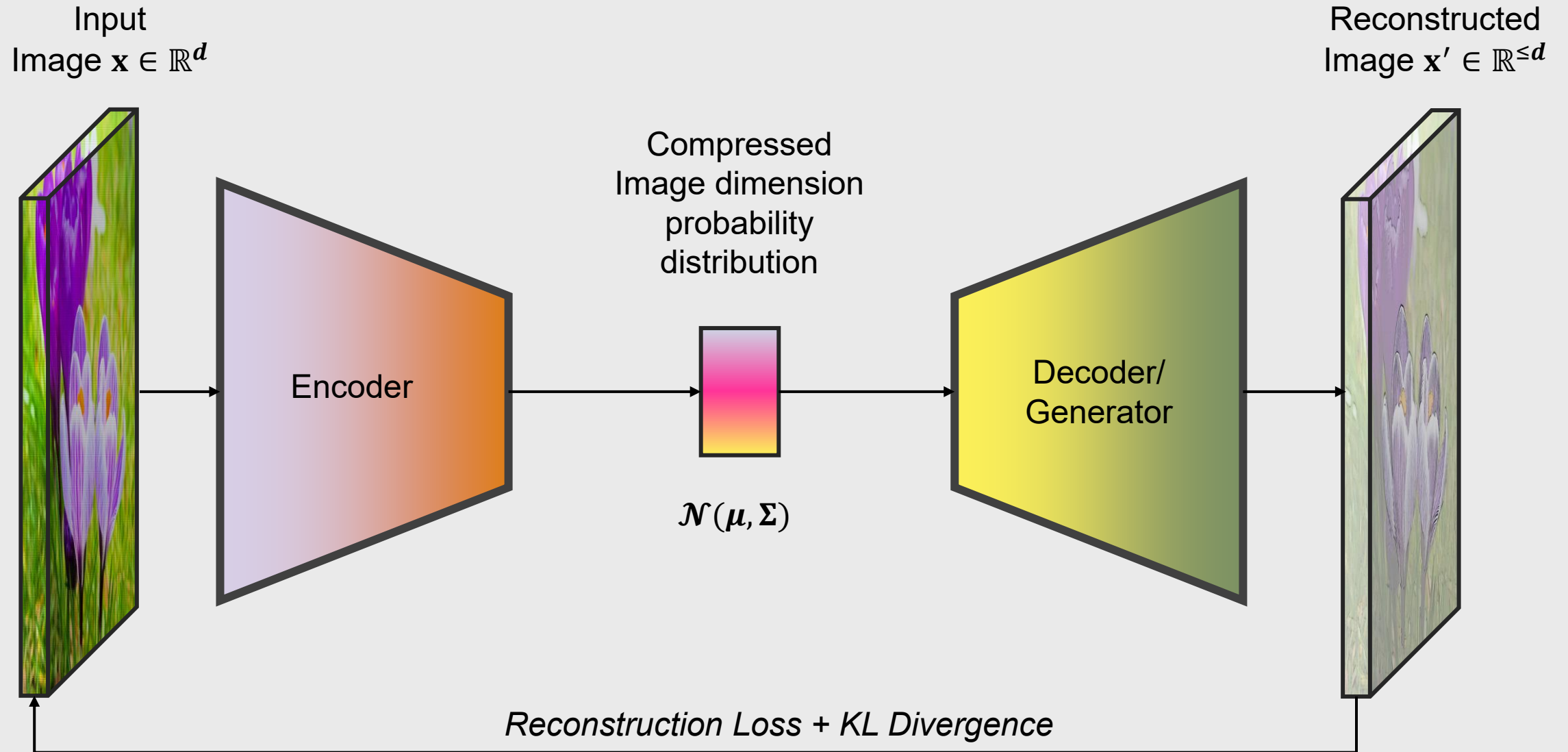
Reconstruction Loss (e.g., Mean Squared Error),
$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (x_i - x'_i)^2$$
 pixel-by-pixel comparison



Latent space give compressed feature representation of input. But it give no guarantee that the hidden units extract meaningful feature.

Variational Autoencoder

9:46 AM



Auto-Encoding Variational Bayes

Diederik P. Kingma

Machine Learning Group
Universiteit van Amsterdam
dpkingma@gmail.com

Max Welling

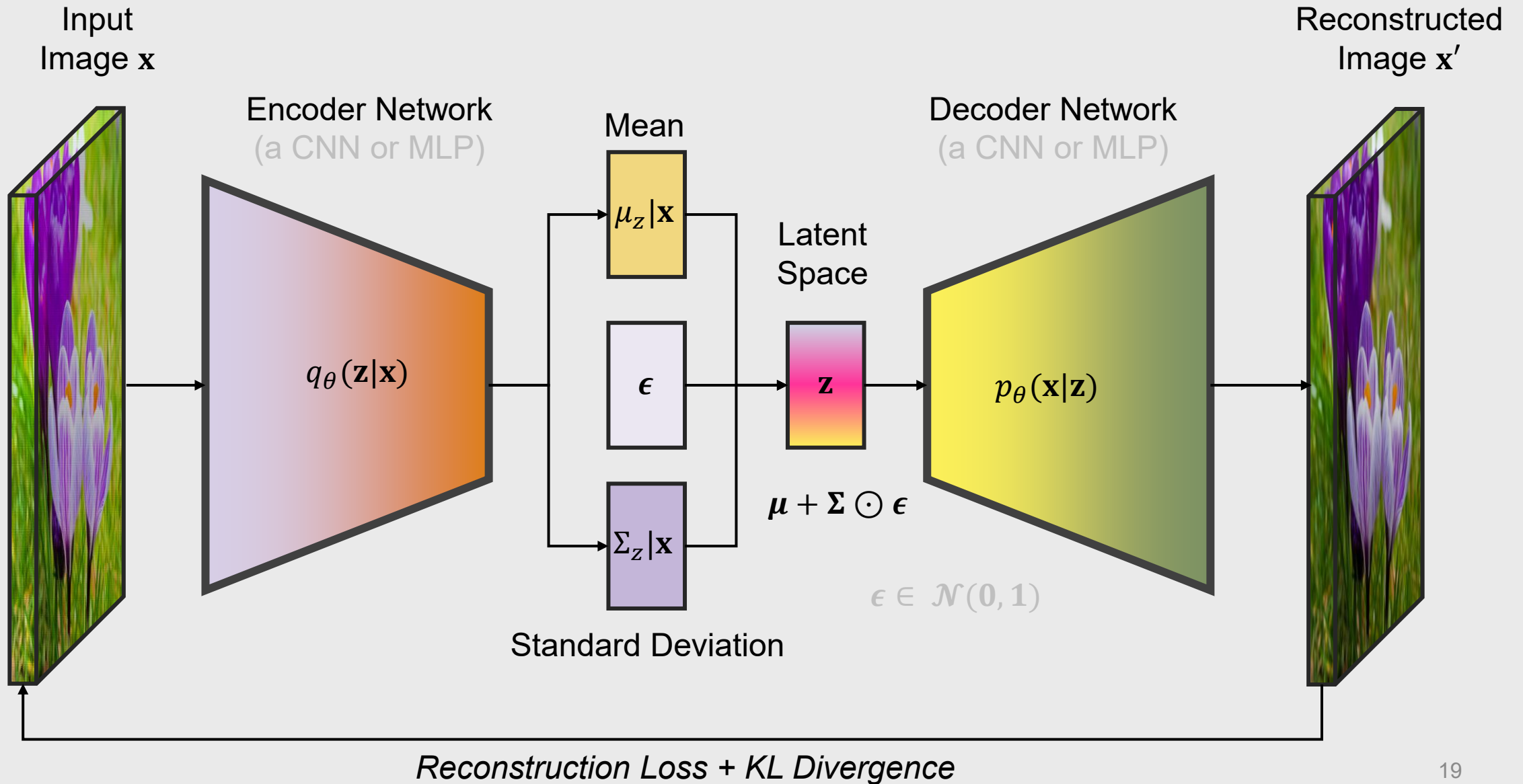
Machine Learning Group
Universiteit van Amsterdam
welling.max@gmail.com

Abstract

How can we perform efficient inference and learning in directed probabilistic models, in the presence of continuous latent variables with intractable posterior distributions, and large datasets? We introduce a stochastic variational inference and learning algorithm that scales to large datasets and, under some mild differentiability conditions, even works in the intractable case. Our contributions is two-fold. First, we show that a reparameterization of the variational lower bound yields a lower bound estimator that can be straightforwardly optimized using standard stochastic gradient methods. Second, we show that for i.i.d. datasets with continuous latent variables per datapoint, posterior inference can be made especially efficient by fitting an approximate inference model (also called a recognition model) to the intractable posterior using the proposed lower bound estimator. Theoretical advantages are reflected in experimental results.

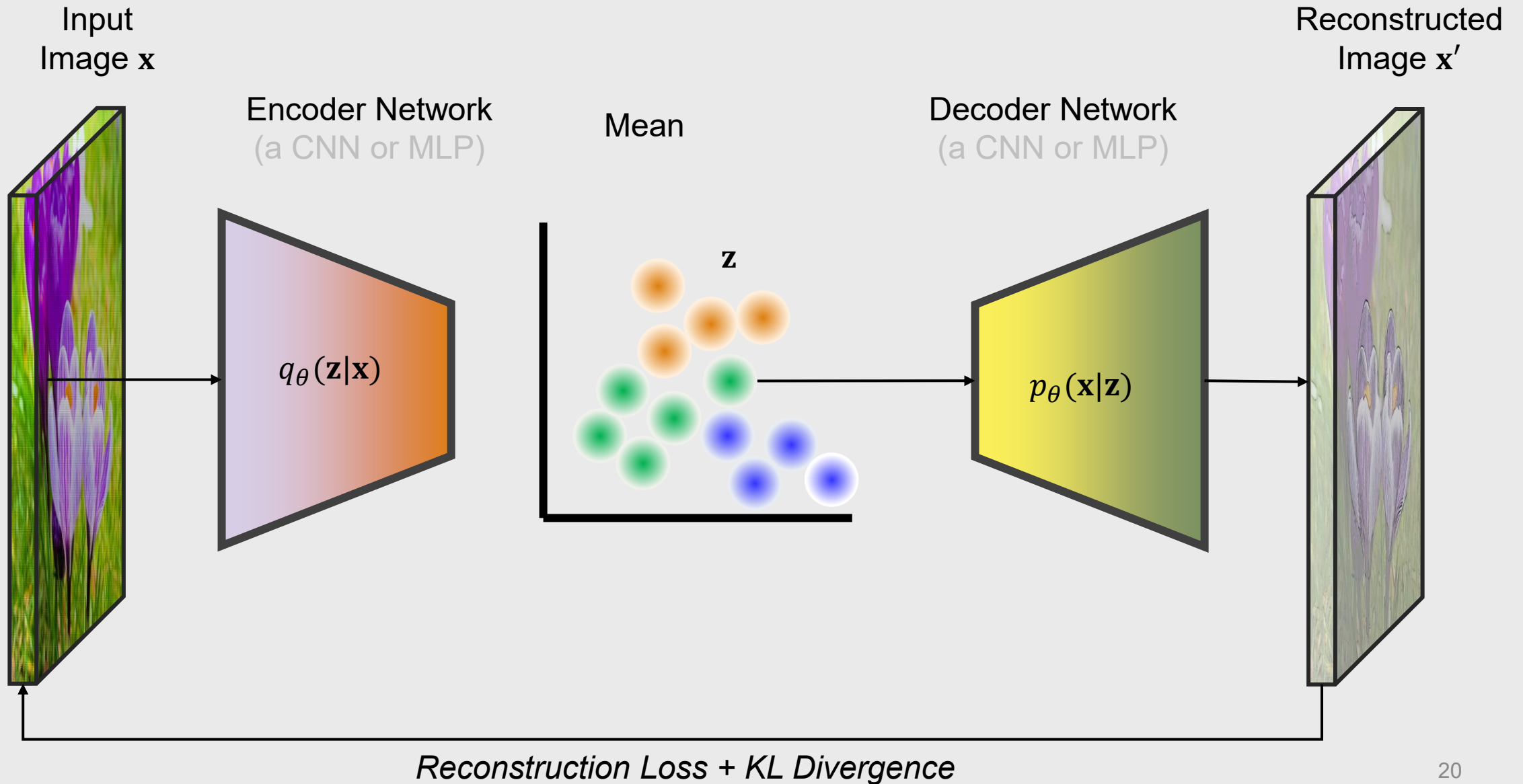
Variational Autoencoder

9:46 AM



Variational Autoencoder

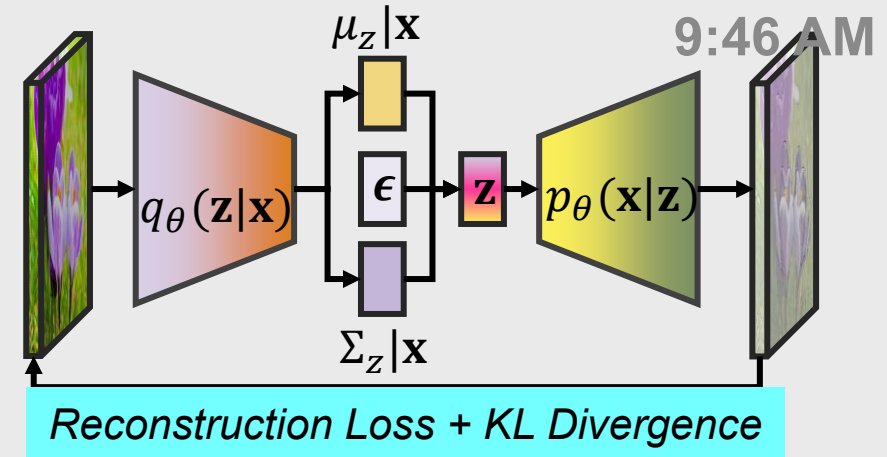
9:46 AM



Loss function of VAE

VAE Loss = Reconstruction Loss + Regularisation Loss

$$L_{VAE} = L_{Reconstruction} + L_{Regularisation}$$



Reconstruction Loss

It answers how well the decoder reconstructs the input from the latent sample.

- **Mean Squared Error (MSE)** for continuous data and
- **Cross Entropy (CE)** categorical data (e.g., MNIST)

$$L_{Reconstruction} = \mathbb{E}_{q(z|x)} [-\log p(x|z)]$$

Regularization Loss

It answers how far the learned latent distribution $q(z|x)$ is from the desired prior distribution $p(z) = N(0, 1)$

$$L_{Regularisation} = D_{KL}(q(z|x) || p(z))$$

Mean Squared Error

5	3
1	6

Image IN
A

2	4
1	3

Image Gen
B

$(5 - 2)^2$	$(3 - 4)^2$
$(1 - 1)^2$	$(6 - 3)^2$

Pixelwise squared difference

9 + 1 +
0 + 9 =

Pixelwise sum

4.75

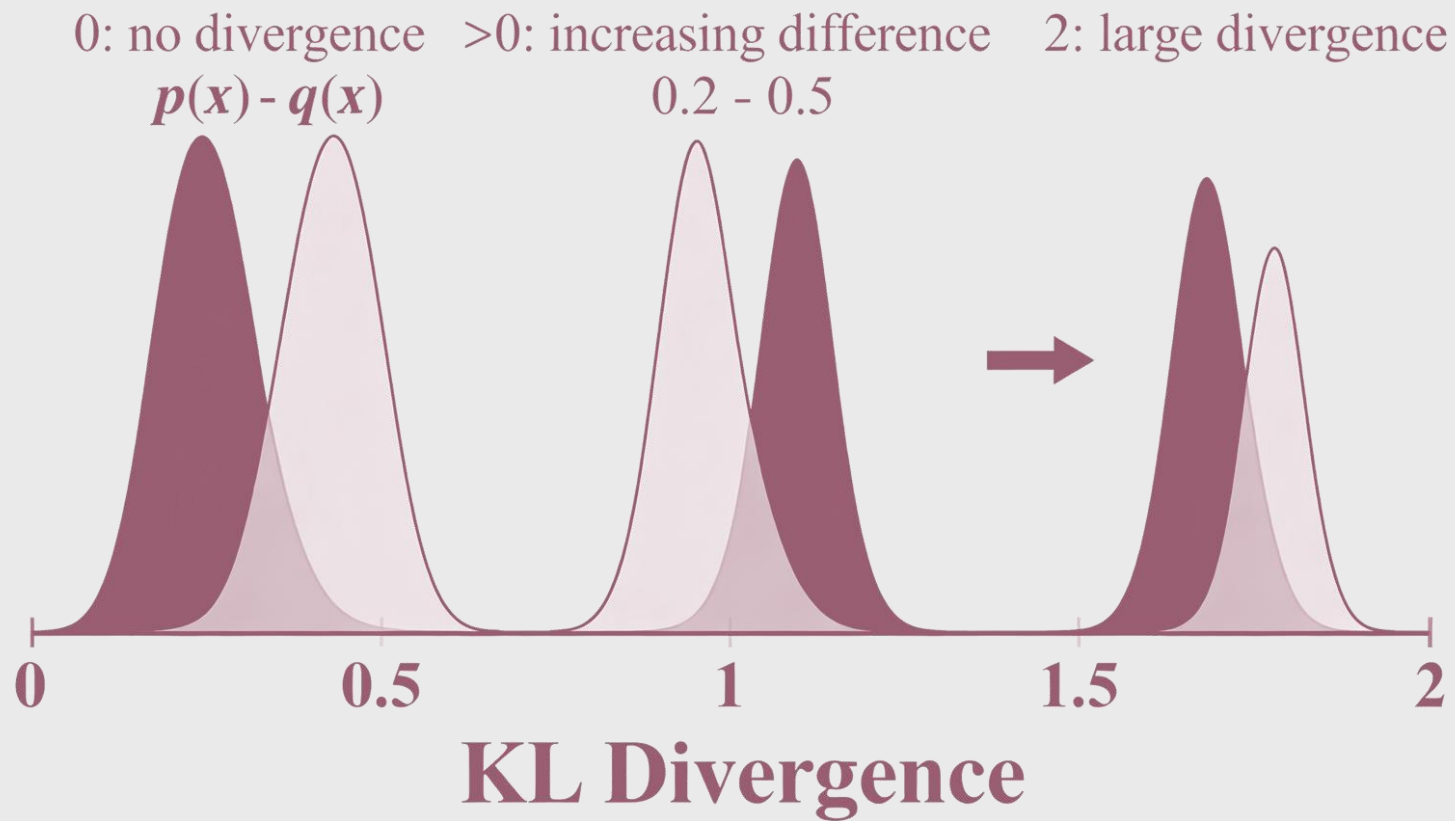
Mean

$$MSE = \frac{1}{N} \sum_{ij}^N (A_{ij} - B_{ij})^2$$

where N is the dimension of a square Metrix/Image

Kullback–Leibler (KL) divergence

KL Divergence measure distance between two probability distributions (much like distance between two points)



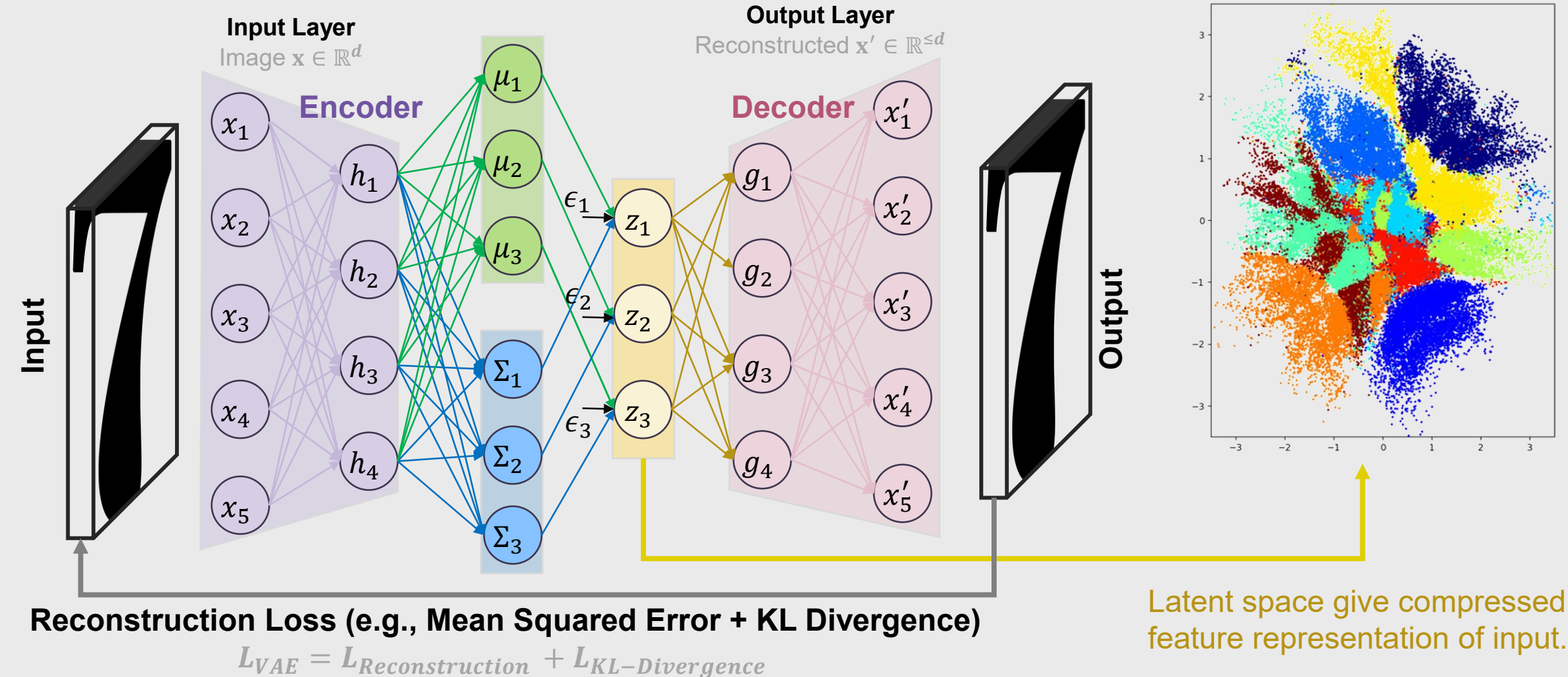
$$D_{KL} = \sum_{x \in X} p(x) \log \left(\frac{p(x)}{q(x)} \right)$$

The lower bound value is zero and is achieved when the distributions under observation are identical.

Variational Autoencoder

9:46 AM

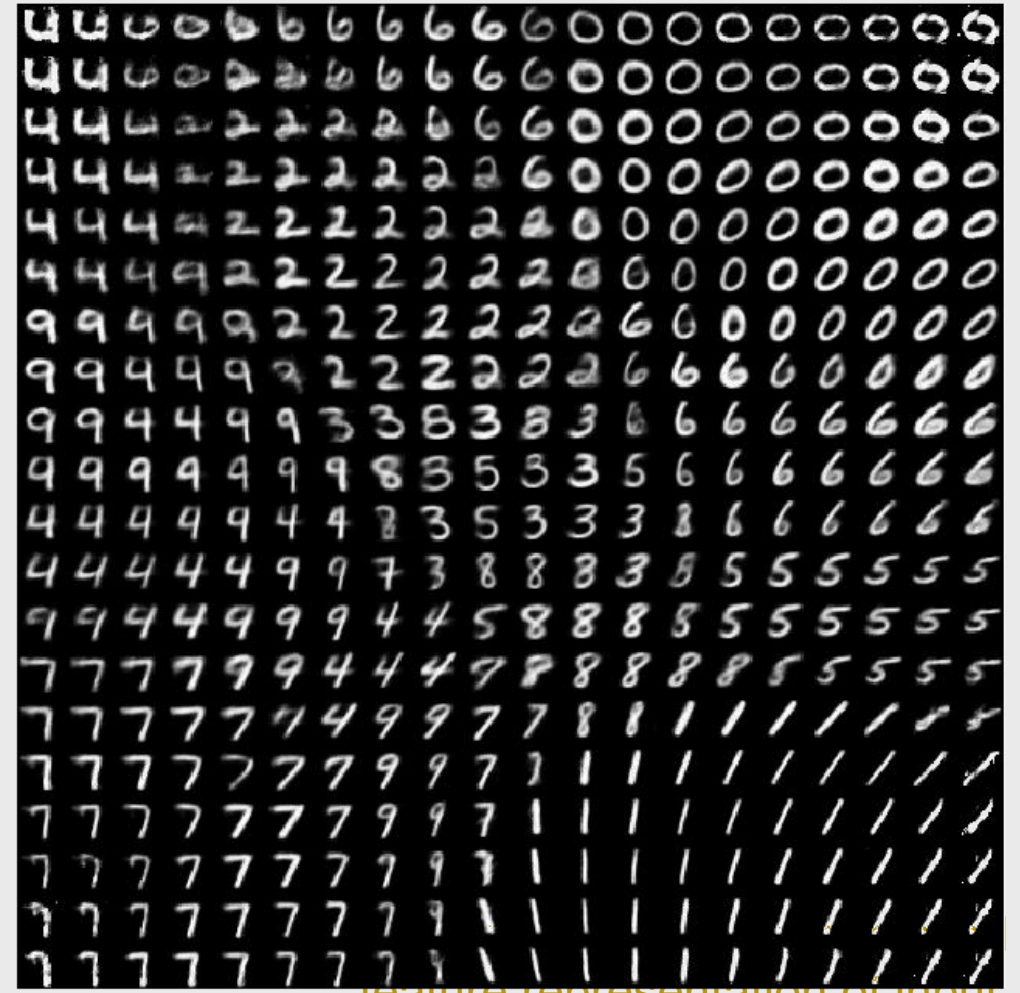
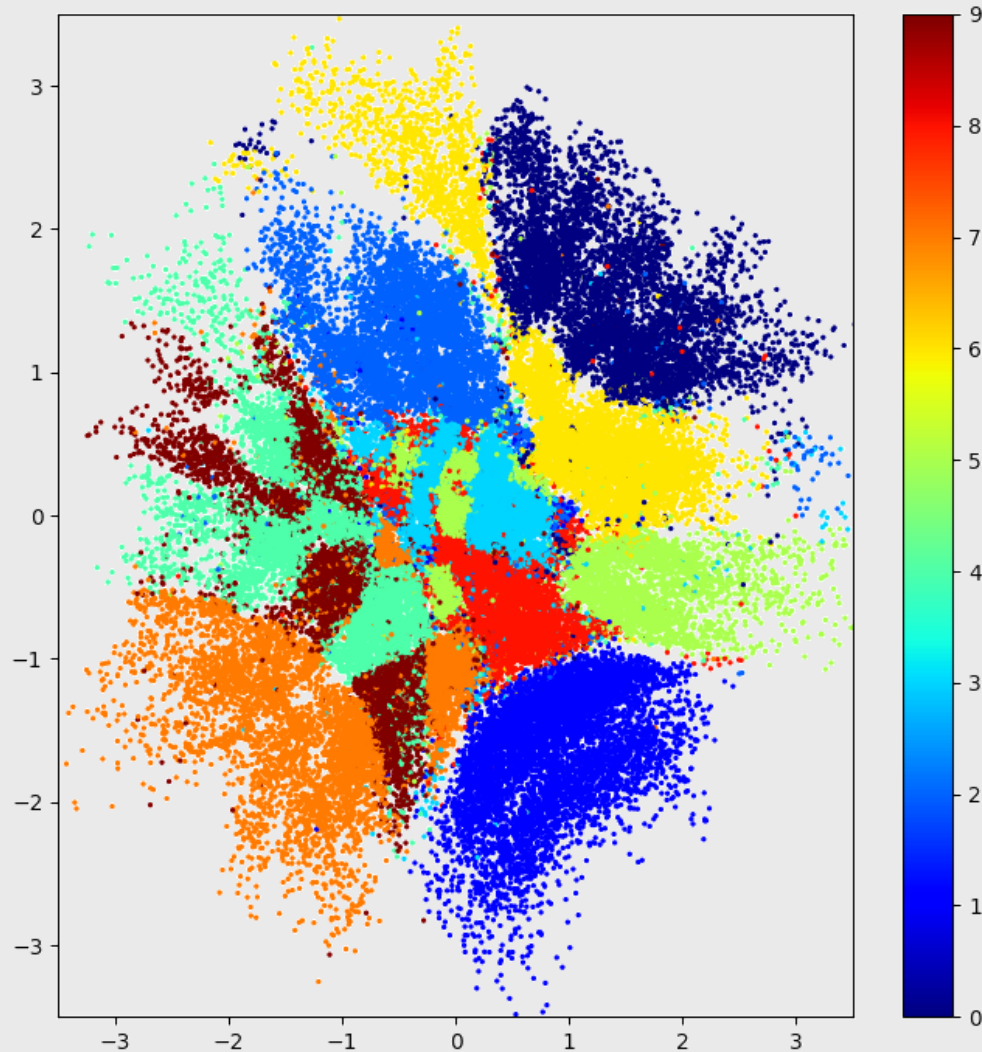
Autoencoder takes its feedback/error correction signal from the input itself. This is **Unsupervised Or Self-Supervised**



Variational Autoencoder

9:46 AM

Autoencoder takes its feedback/error correction signal from the input itself. This is **Unsupervised Or Self-Supervised**

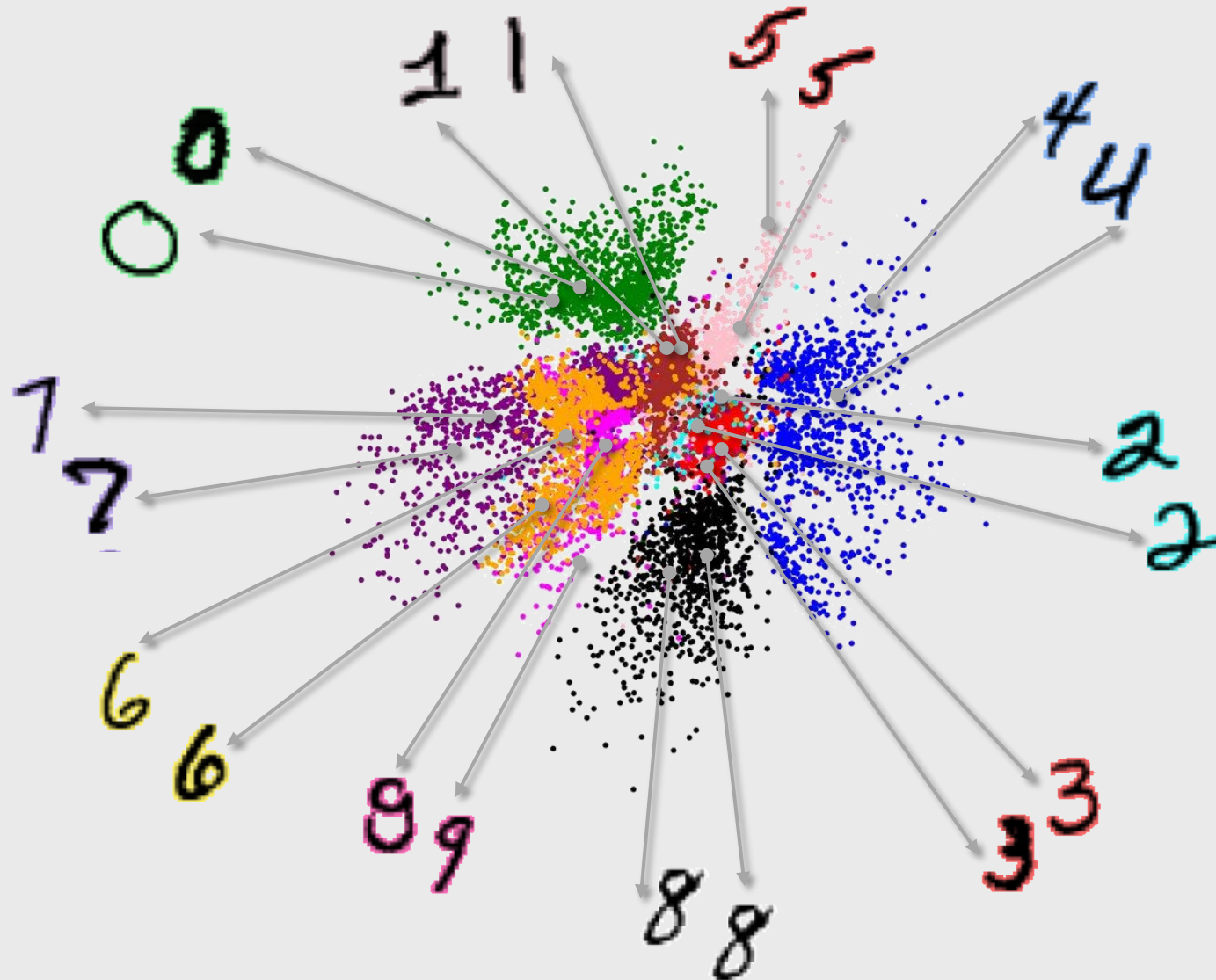


feature representation of input.

Variational Autoencoder

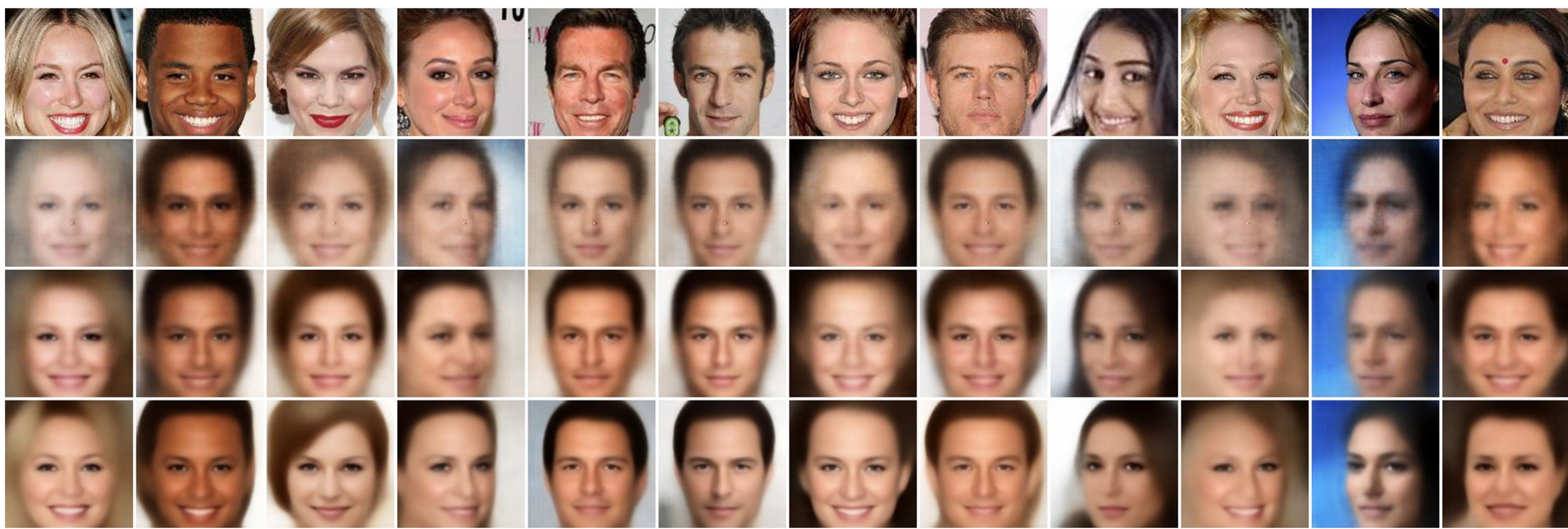
9:46 AM

Latent space give compressed feature representation of input.



Reconstructed Images of Celeb Face Image

9:46 AM



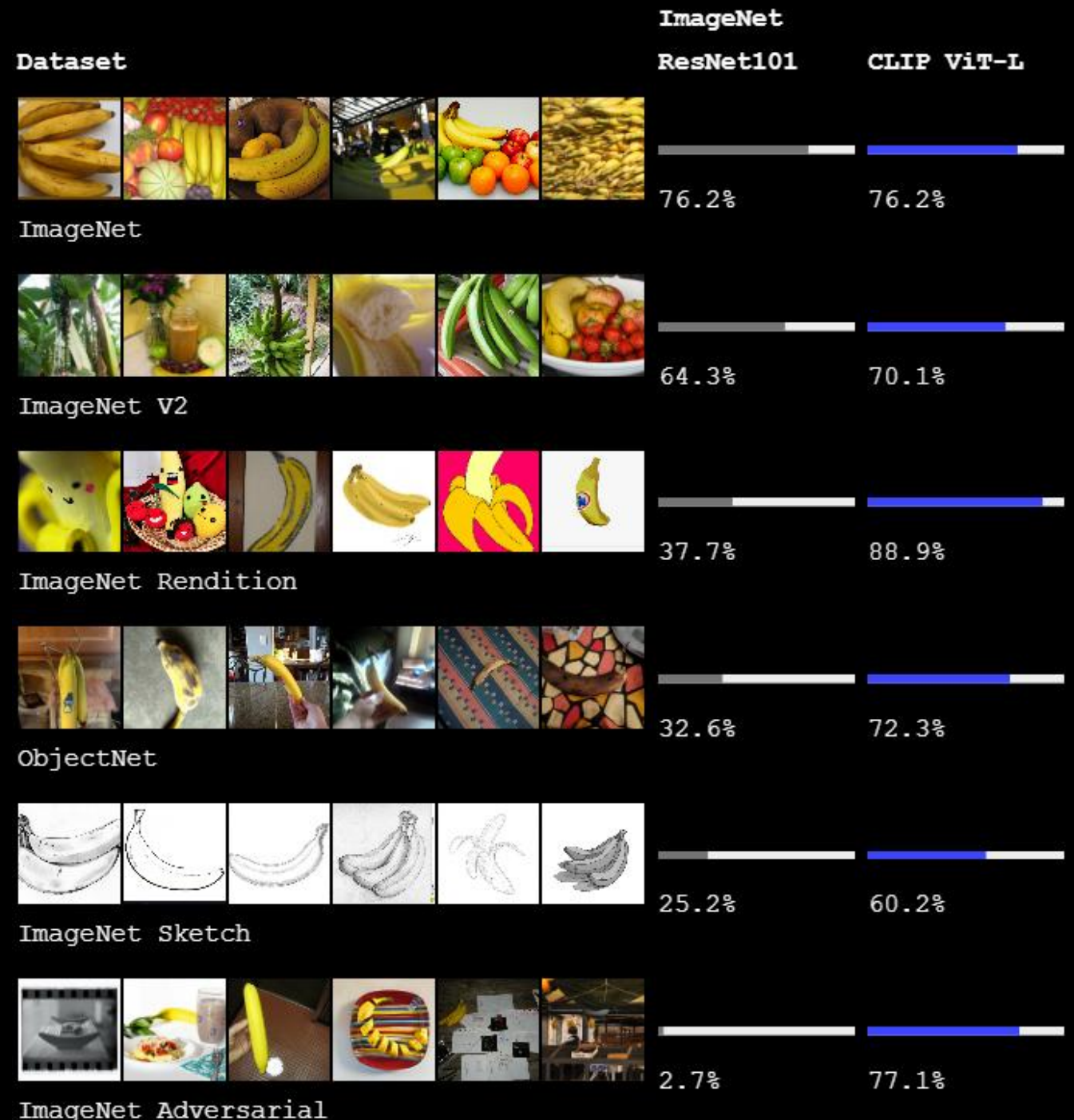
Cai et al (2017) Multi-Stage Variational Auto-Encoders for Coarse-to-Fine Image Generation: <https://arxiv.org/abs/1705.07202>

CLIP Models

Contrastive Language-Image Pre-training

- CLIP efficiently learns visual concepts from natural language supervision.
- It uses Image and Text pairs as training data.
- It has Zero-Shot classifier capabilities, meaning it recognises image/objects just by name or text prompt.
- Unlike traditional image classifier, it fare higher robustness.

Source: <https://openai.com/index/clip/>



Alec Radford^{*1} Jong Wook Kim^{*1} Chris Hallacy¹ Aditya Ramesh¹ Gabriel Goh¹ Sandhini Agarwal¹
Girish Sastry¹ Amanda Askell¹ Pamela Mishkin¹ Jack Clark¹ Gretchen Krueger¹ Ilya Sutskever¹

Abstract

State-of-the-art computer vision systems are trained to predict a fixed set of predetermined object categories. This restricted form of supervision limits their generality and usability since additional labeled data is needed to specify any other visual concept. Learning directly from raw text about images is a promising alternative which leverages a much broader source of supervision. We demonstrate that the simple pre-training task of predicting which caption goes with which image is an efficient and scalable way to learn SOTA image representations from scratch on a dataset of 400 million (image, text) pairs collected from the internet. After pre-training, natural language is used to reference learned visual concepts (or describe new ones) enabling zero-shot transfer of the model to downstream tasks. We study the performance of this approach by benchmarking on over 30 different existing computer vision datasets, spanning tasks such as OCR, action recognition in videos, geo-localization, and many types of fine-grained object classification. The model transfers non-trivially to most tasks and is often competitive with a fully supervised baseline without the need for any dataset specific training. For instance, we match the accuracy of the original ResNet-50 on ImageNet zero-shot without needing to use any of the 1.28 million training examples it was trained on. We release our code and pre-trained model weights at <https://github.com/OpenAI/CLIP>.

1. Introduction and Motivating Work

Pre-training methods which learn directly from raw text have revolutionized NLP over the last few years (Dai & Le, 2015; Peters et al., 2018; Howard & Ruder, 2018; Radford et al., 2018; Devlin et al., 2018; Raffel et al., 2019).

^{*}Equal contribution ¹OpenAI, San Francisco, CA 94110, USA. Correspondence to: <{alec, jongwook}@openai.com>.

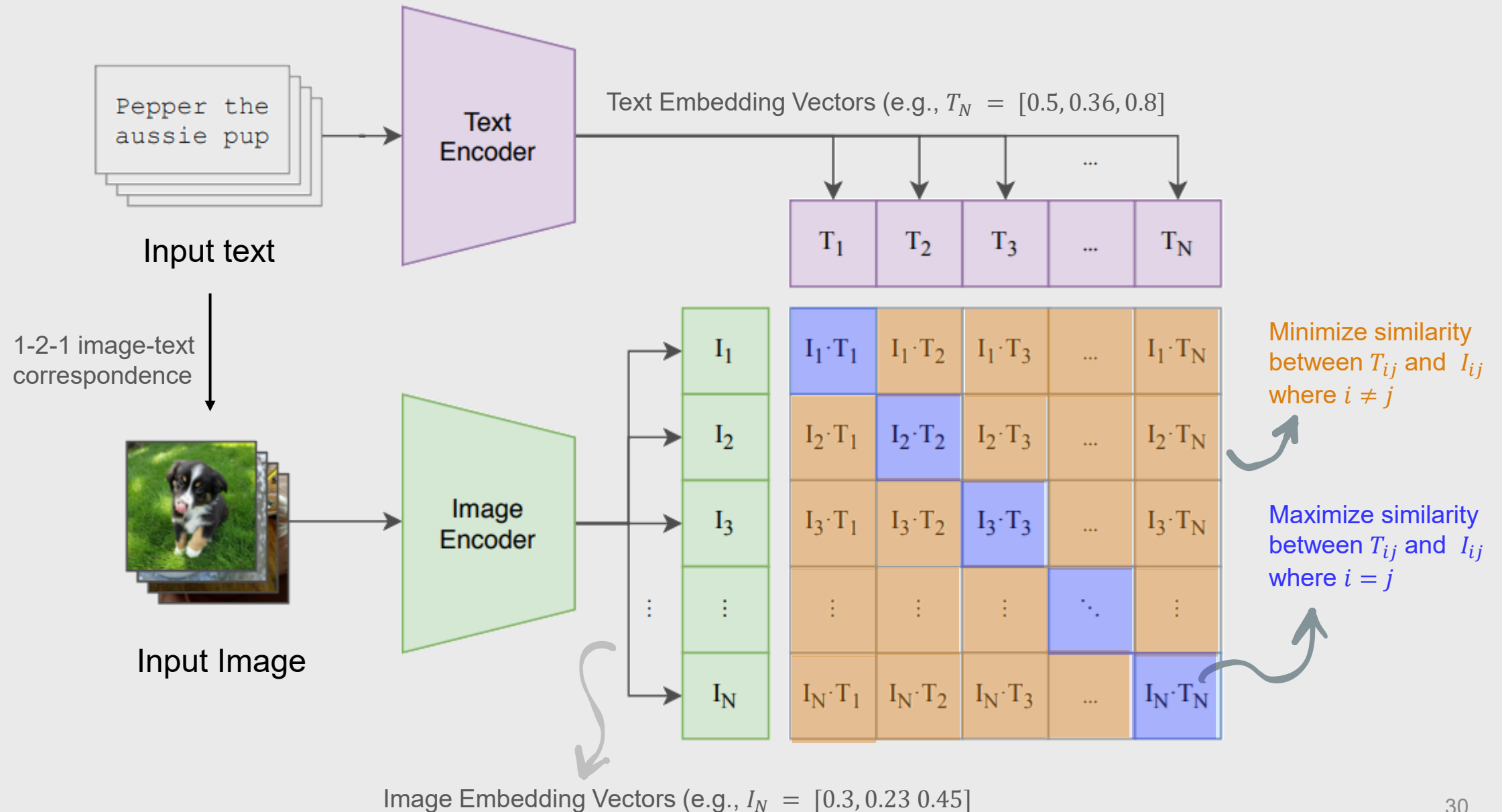
Task-agnostic objectives such as autoregressive and masked language modeling have scaled across many orders of magnitude in compute, model capacity, and data, steadily improving capabilities. The development of “text-to-text” as a standardized input-output interface (McCann et al., 2018; Radford et al., 2019; Raffel et al., 2019) has enabled task-agnostic architectures to zero-shot transfer to downstream datasets removing the need for specialized output heads or dataset specific customization. Flagship systems like GPT-3 (Brown et al., 2020) are now competitive across many tasks with bespoke models while requiring little to no dataset specific training data.

These results suggest that the aggregate supervision accessible to modern pre-training methods within web-scale collections of text surpasses that of high-quality crowd-labeled NLP datasets. However, in other fields such as computer vision it is still standard practice to pre-train models on crowd-labeled datasets such as ImageNet (Deng et al., 2009). Could scalable pre-training methods which learn directly from web text result in a similar breakthrough in computer vision? Prior work is encouraging.

Over 20 years ago Mori et al. (1999) explored improving content based image retrieval by training a model to predict the nouns and adjectives in text documents paired with images. Quattoni et al. (2007) demonstrated it was possible to learn more data efficient image representations via manifold learning in the weight space of classifiers trained to predict words in captions associated with images. Srivastava & Salakhutdinov (2012) explored deep representation learning by training multimodal Deep Boltzmann Machines on top of low-level image and text tag features. Joulin et al. (2016) modernized this line of work and demonstrated that CNNs trained to predict words in image captions learn useful image representations. They converted the title, description, and hashtag metadata of images in the YFCC100M dataset (Thomee et al., 2016) into a bag-of-words multi-label classification task and showed that pre-training AlexNet (Krizhevsky et al., 2012) to predict these labels learned representations which preformed similarly to ImageNet-based pre-training on transfer tasks. Li et al. (2017) then extended this approach to predicting phrase n-grams in addition to individual words and demonstrated the ability of their system to zero-shot transfer to other image

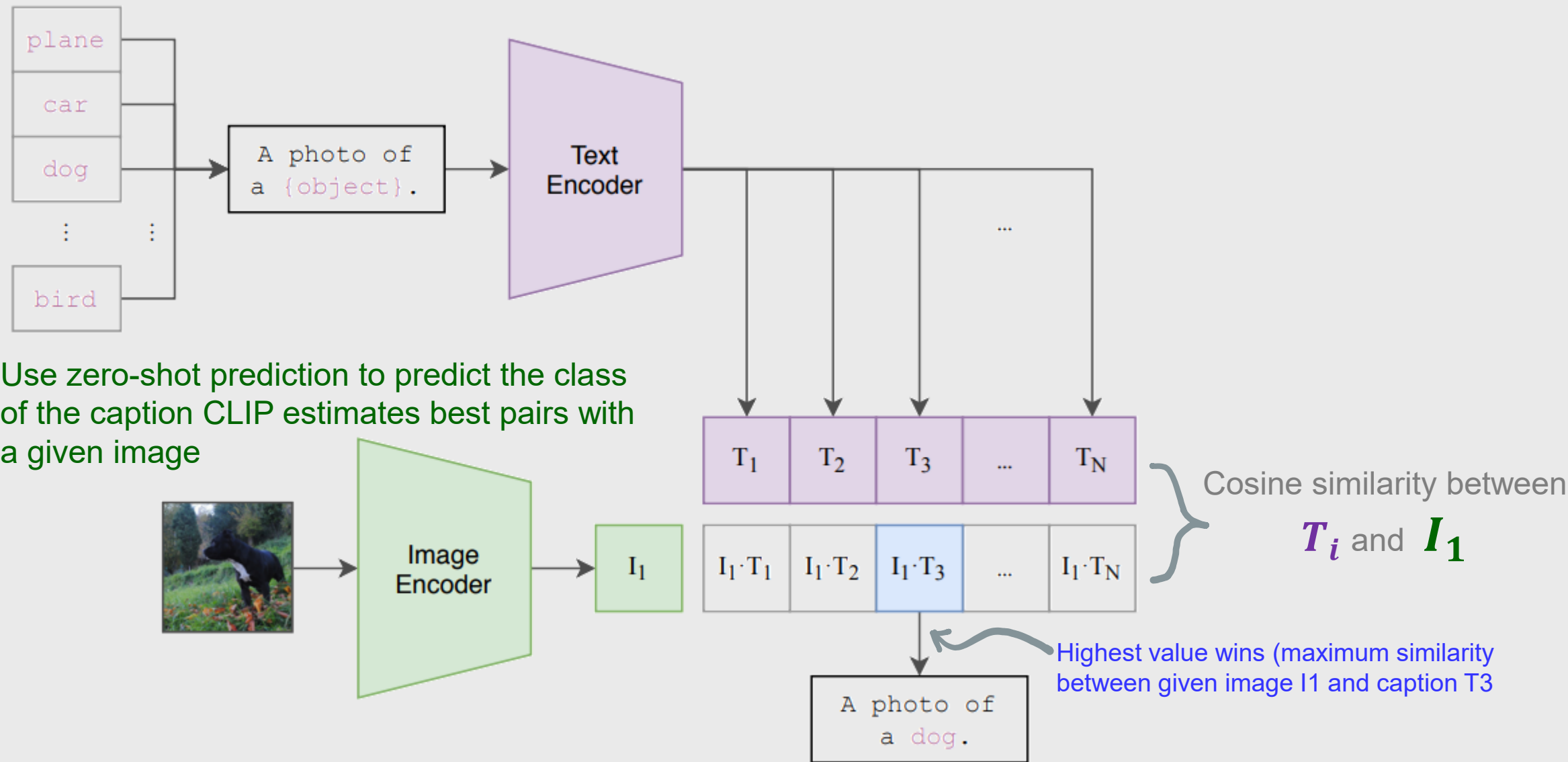
CLIP pre-trains an image encoder and a text encoder
to predict which images were paired with which texts in our dataset

9:46 AM

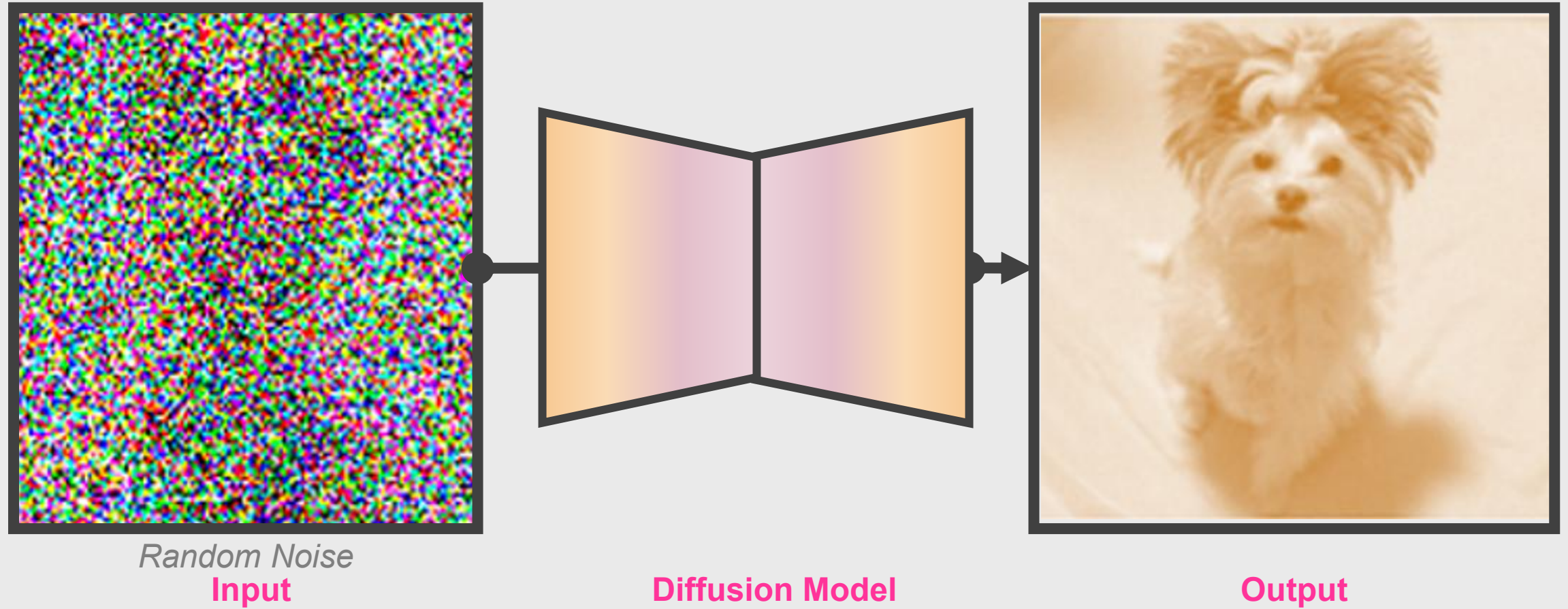


Convert all of a dataset's classes into captions such as "a photo of a dog"

9:46 AM



Diffusion Models



Diffusion

9:46 AM

random molecular movement cause molecule to spread out all over, or *diffuse*, over space

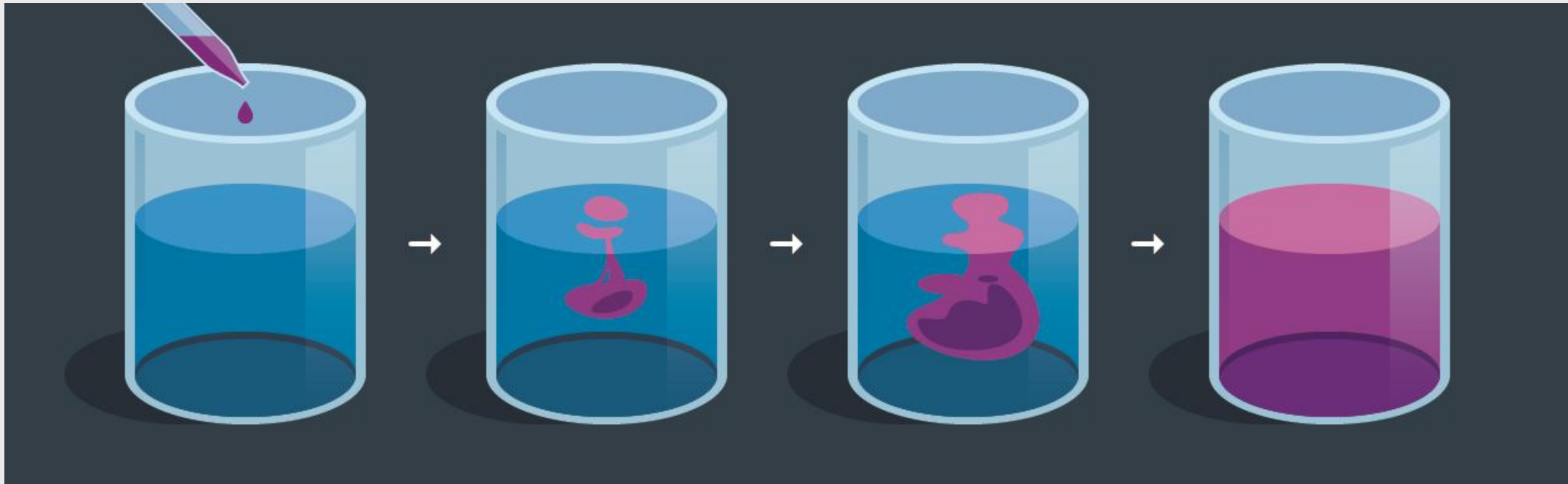
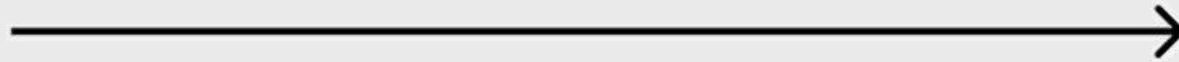


image credit: Visionlearning

Diffusion Models

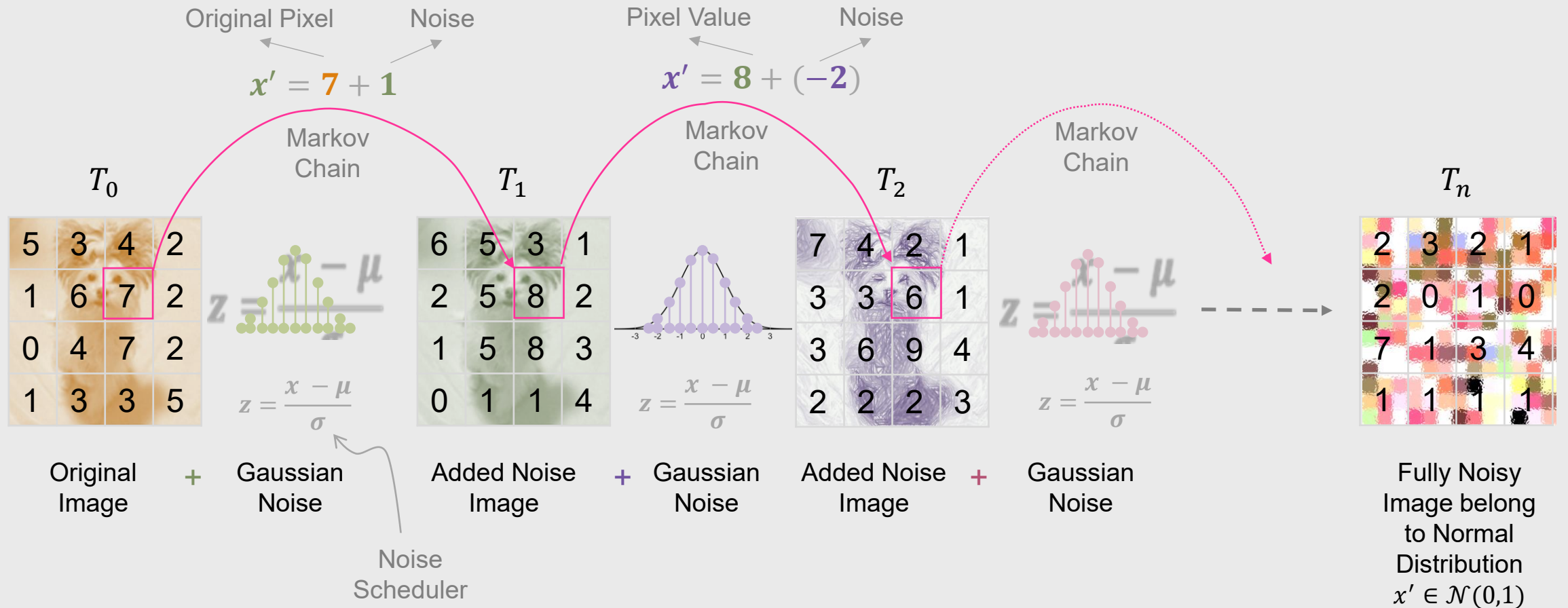
forward diffusion



reverse diffusion

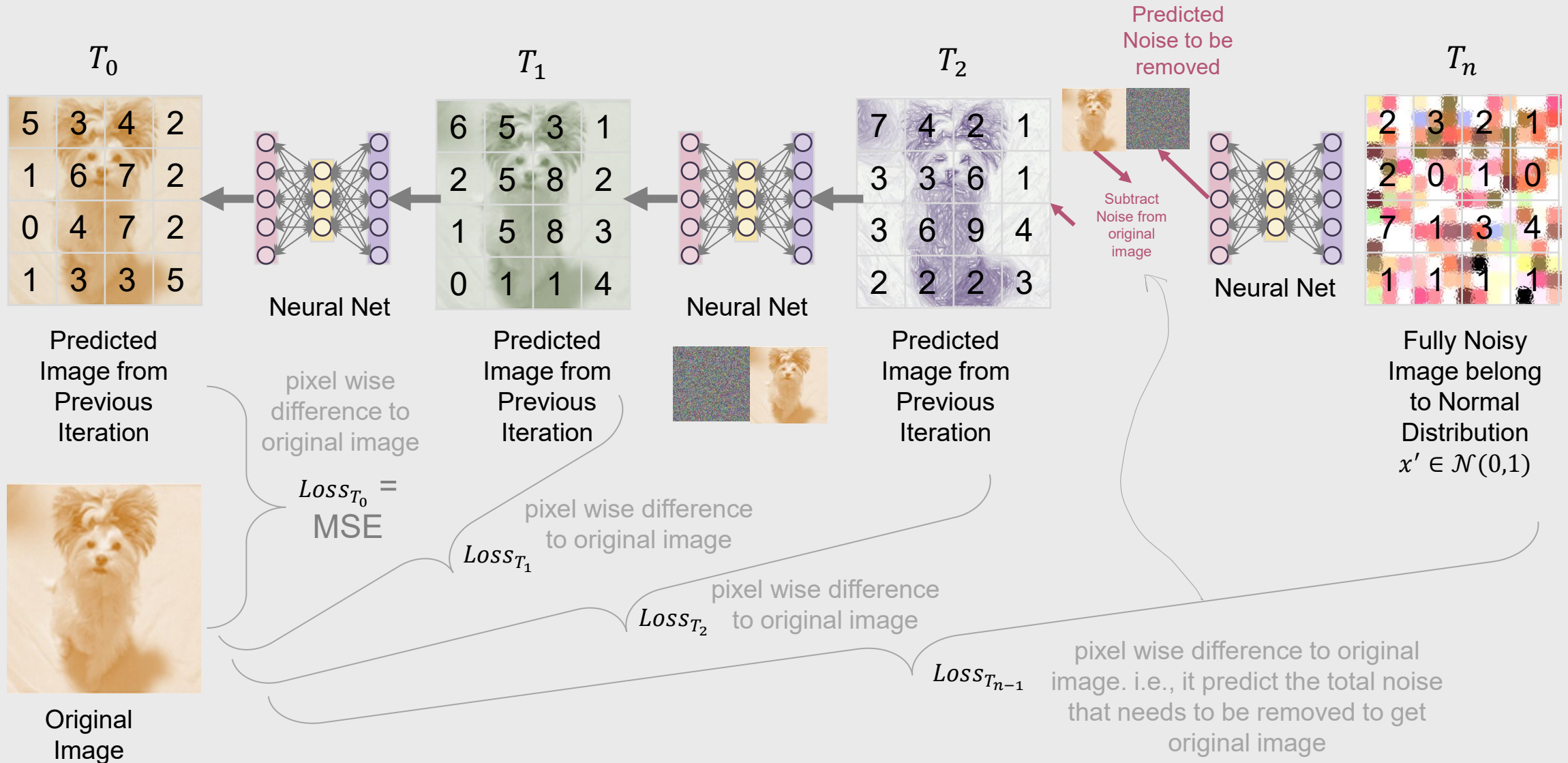
Forward Diffusion

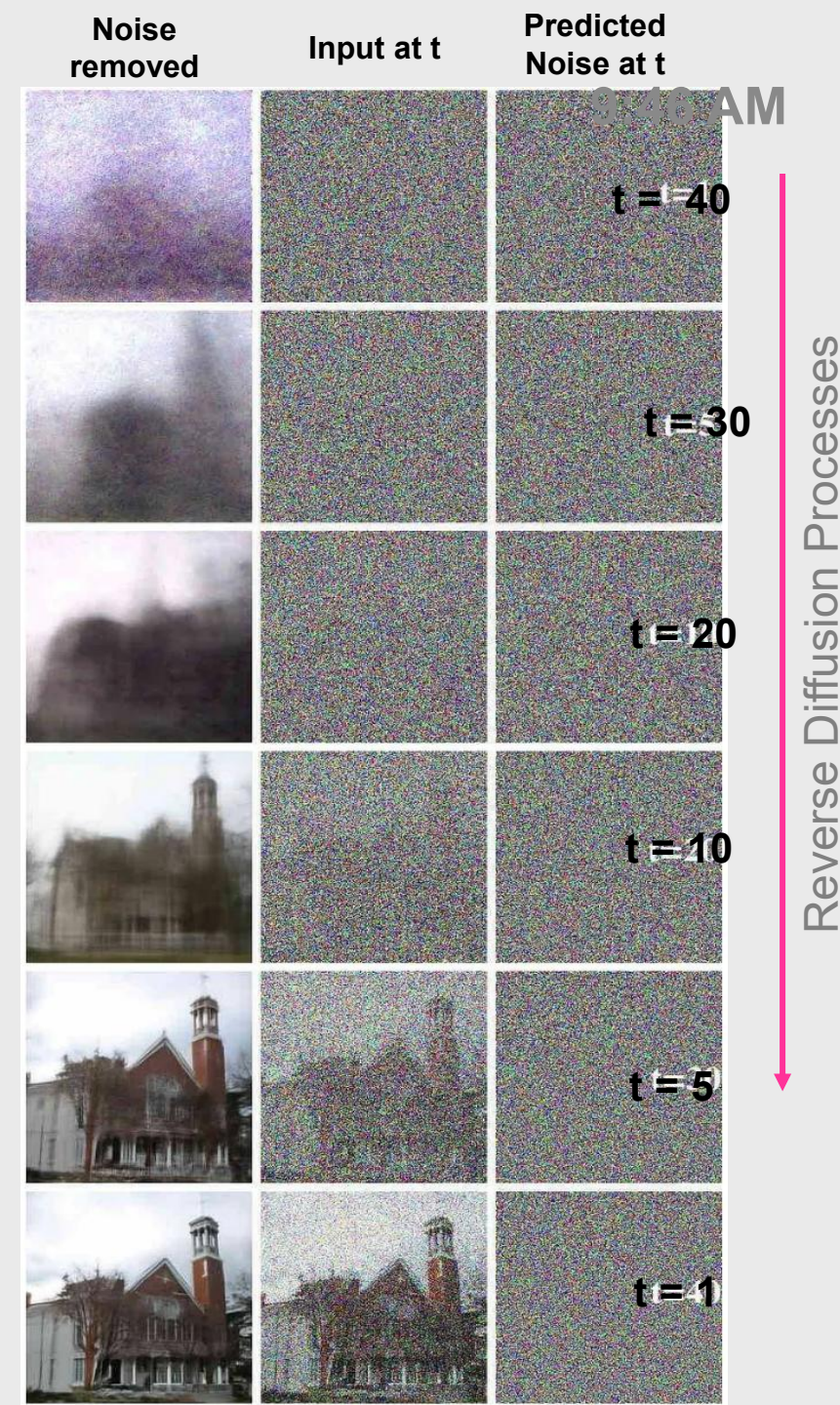
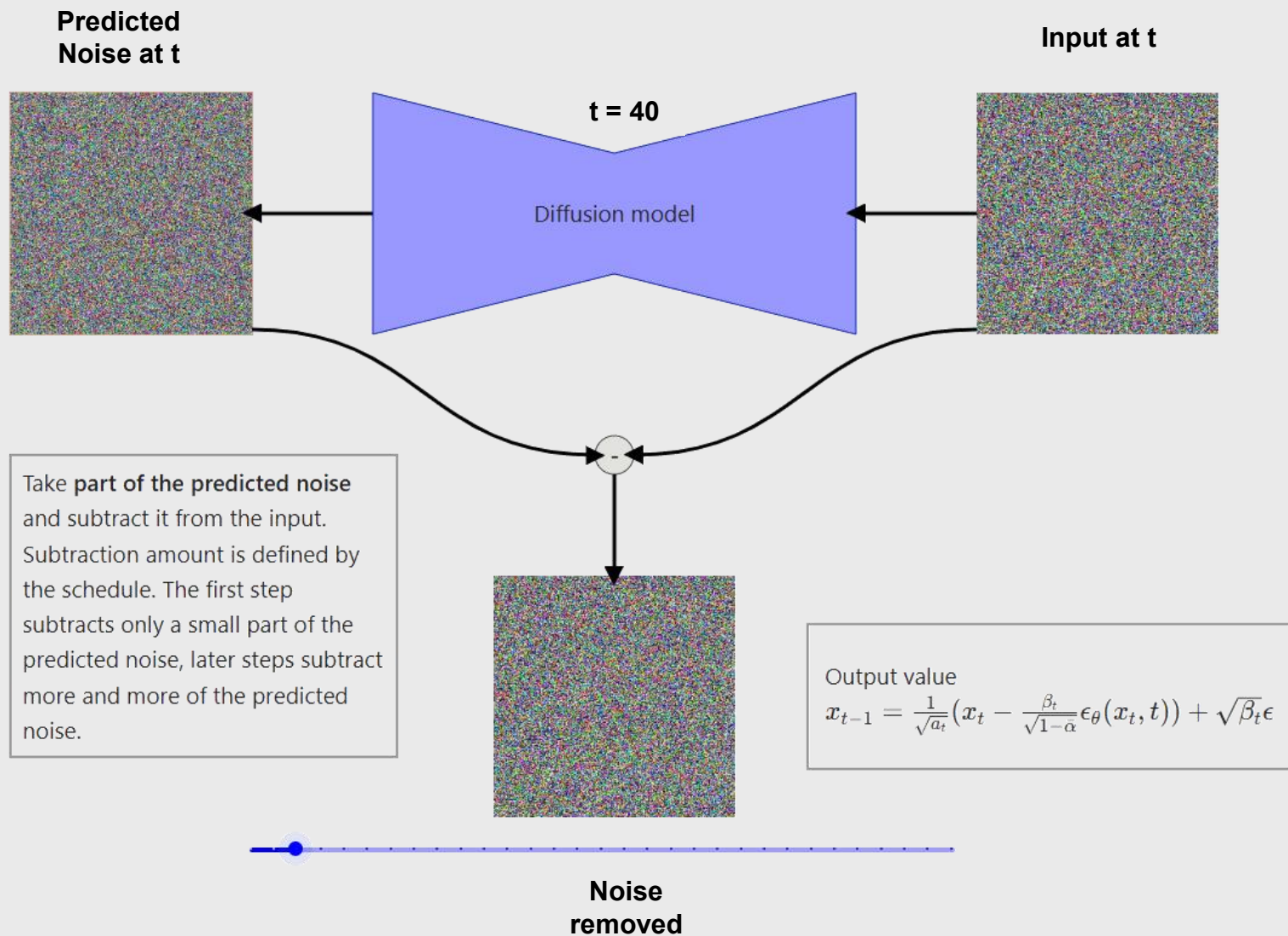
9:46 AM



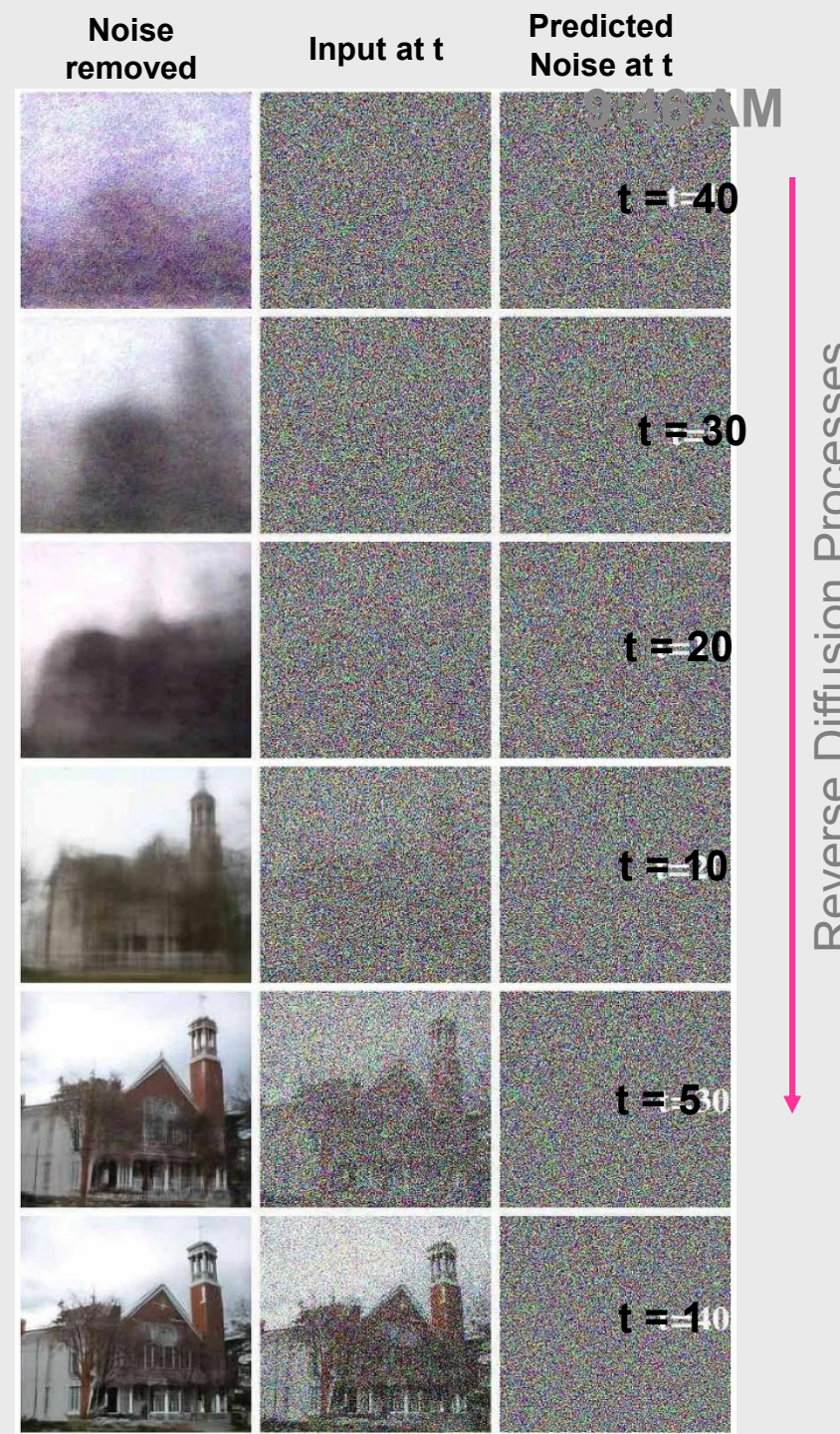
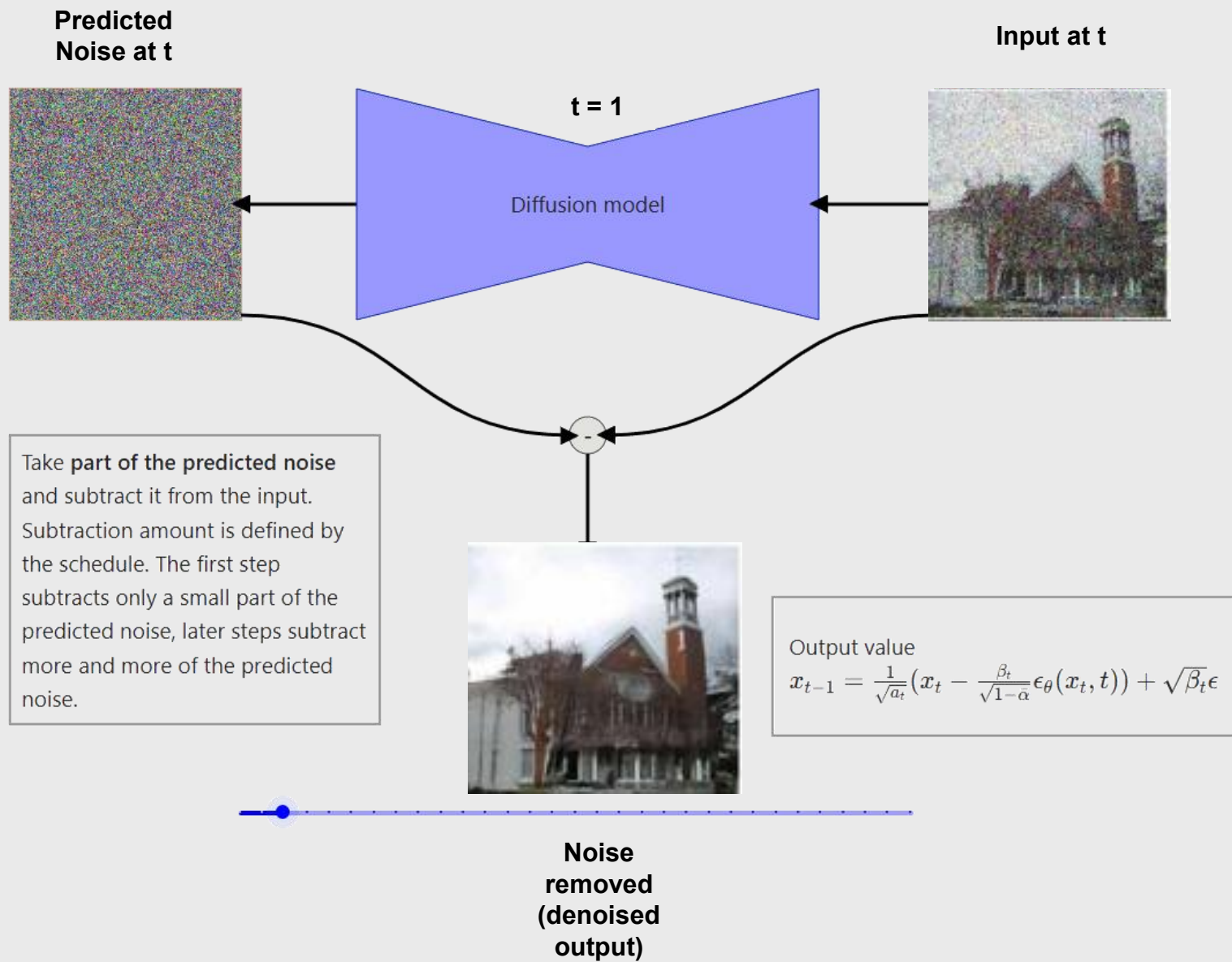
Reverse Diffusion

9:46 AM





Source & Visual Illustration: <https://erdem.pl/2023/11/step-by-step-visual-introduction-to-diffusion-models#reverse-diffusion-diagram>

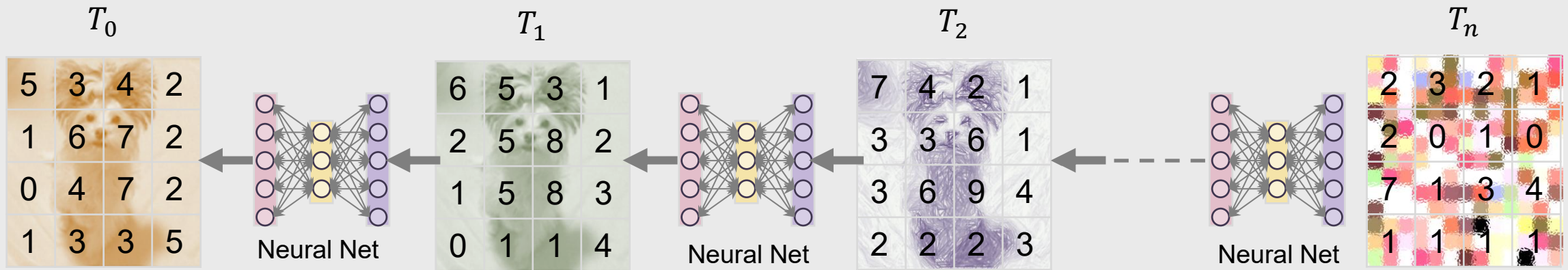


Source & Visual Illustration: <https://erdem.pl/2023/11/step-by-step-visual-introduction-to-diffusion-models#reverse-diffusion-diagram>

Reserve Diffusion

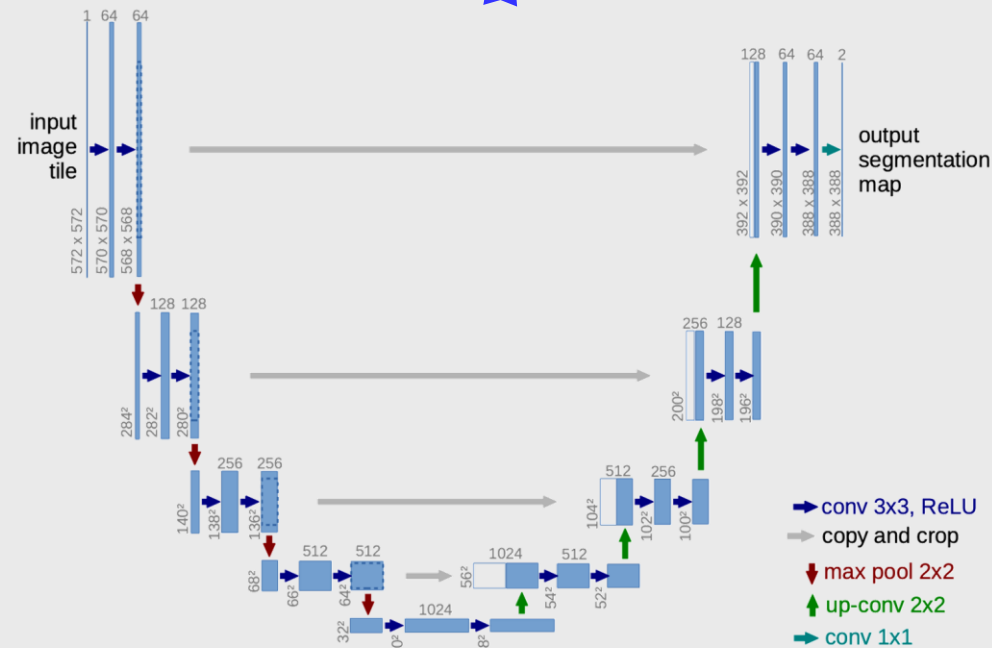
9:46 AM

A U-Net architecture is used as the backbone in diffusion models



Shared Network

A U-Net architecture is used as the backbone in diffusion models. A single shared U-Net is trained on successive pairs of images (T_i, T_{i-1}) at different diffusion steps



We train net to minimize this loss function

$$\mathbb{E}_{\mathbf{x}_0, \epsilon} \left[\frac{\beta_t^2}{2\sigma_t^2 \alpha_t (1 - \bar{\alpha}_t)} \left\| \epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t) \right\|^2 \right]$$

Jonathan Ho UC Berkeley jonathanho@berkeley.edu
 Ajay Jain UC Berkeley ajayj@berkeley.edu
 Pieter Abbeel UC Berkeley pabbeel@cs.berkeley.edu

Abstract

We present high quality image synthesis results using diffusion probabilistic models, a class of latent variable models inspired by considerations from nonequilibrium thermodynamics. Our best results are obtained by training on a weighted variational bound designed according to a novel connection between diffusion probabilistic models and denoising score matching with Langevin dynamics, and our models naturally admit a progressive lossy decompression scheme that can be interpreted as a generalization of autoregressive decoding. On the unconditional CIFAR10 dataset, we obtain an Inception score of 9.46 and a state-of-the-art FID score of 3.17. On 256x256 LSUN, we obtain sample quality similar to ProgressiveGAN. Our implementation is available at <https://github.com/hojonathanho/diffusion>.

1 Introduction

Deep generative models of all kinds have recently exhibited high quality samples in a wide variety of data modalities. Generative adversarial networks (GANs), autoregressive models, flows, and variational autoencoders (VAEs) have synthesized striking image and audio samples [14, 27, 3, 58, 38, 25, 10, 32, 44, 57, 26, 33, 45], and there have been remarkable advances in energy-based modeling and score matching that have produced images comparable to those of GANs [11, 55].

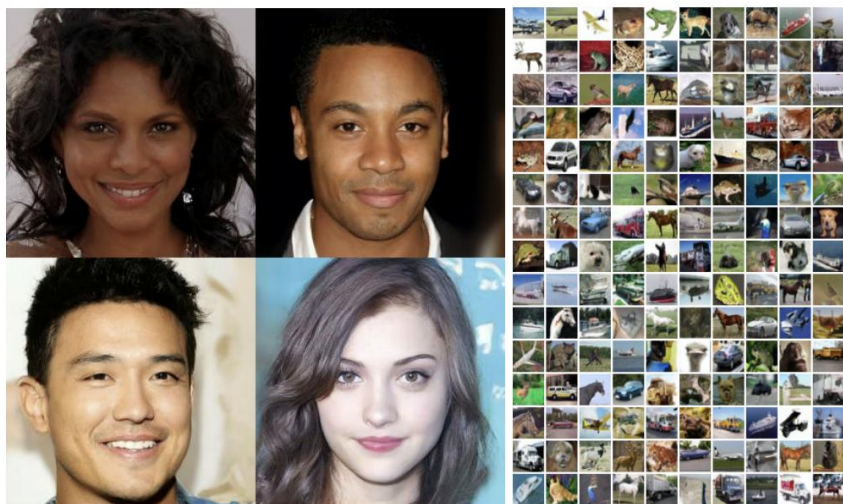


Figure 1: Generated samples on CelebA-HQ 256×256 (left) and unconditional CIFAR10 (right)

Jascha Sohl-Dickstein
 Stanford University

JASCHA@STANFORD.EDU

Eric A. Weiss
 University of California, Berkeley

EWEISS@BERKELEY.EDU

Niru Maheswaranathan
 Stanford University

NIRUM@STANFORD.EDU

Surya Ganguli
 Stanford University

SGANGULI@STANFORD.EDU

Abstract

A central problem in machine learning involves modeling complex data-sets using highly flexible families of probability distributions in which learning, sampling, inference, and evaluation are still analytically or computationally tractable. Here, we develop an approach that simultaneously achieves both flexibility and tractability. The essential idea, inspired by non-equilibrium statistical physics, is to systematically and slowly destroy structure in a data distribution through an iterative forward diffusion process. We then learn a reverse diffusion process that restores structure in data, yielding a highly flexible and tractable generative model of the data. This approach allows us to rapidly learn, sample from, and evaluate probabilities in deep generative models with thousands of layers or time steps, as well as to compute conditional and posterior probabilities under the learned model. We additionally release an open source reference implementation of the algorithm.

1. Introduction

Historically, probabilistic models suffer from a tradeoff between two conflicting objectives: *tractability* and *flexibility*. Models that are *tractable* can be analytically evaluated and easily fit to data (e.g. a Gaussian or Laplace). However,

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these models are unable to aptly describe structure in rich datasets. On the other hand, models that are *flexible* can be molded to fit structure in arbitrary data. For example, we can define models in terms of any (non-negative) function $\phi(\mathbf{x})$ yielding the flexible distribution $p(\mathbf{x}) = \frac{\phi(\mathbf{x})}{Z}$, where Z is a normalization constant. However, computing this normalization constant is generally intractable. Evaluating, training, or drawing samples from such flexible models typically requires a very expensive Monte Carlo process.

A variety of analytic approximations exist which ameliorate, but do not remove, this tradeoff—for instance mean field theory and its expansions (T, 1982; Tanaka, 1998), variational Bayes (Jordan et al., 1999), contrastive divergence (Welling & Hinton, 2002; Hinton, 2002), minimum probability flow (Sohl-Dickstein et al., 2011b;a), minimum KL contraction (Lyu, 2011), proper scoring rules (Gneiting & Raftery, 2007; Parry et al., 2012), score matching (Hyvärinen, 2005), pseudolikelihood (Besag, 1975), loopy belief propagation (Murphy et al., 1999), and many, many more. Non-parametric methods (Gershman & Blei, 2012) can also be very effective¹.

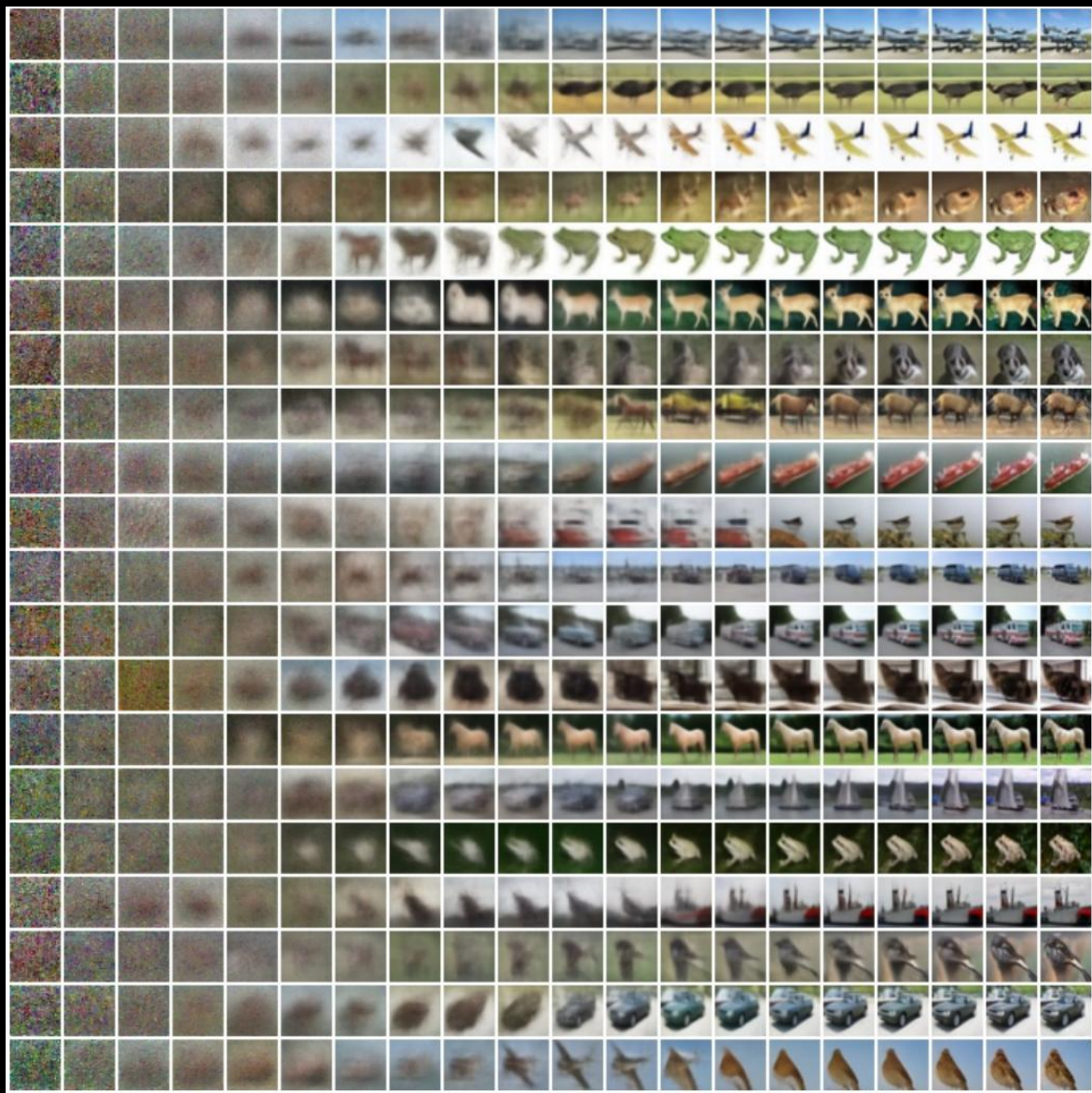
1.1. Diffusion probabilistic models

We present a novel way to define probabilistic models that allows:

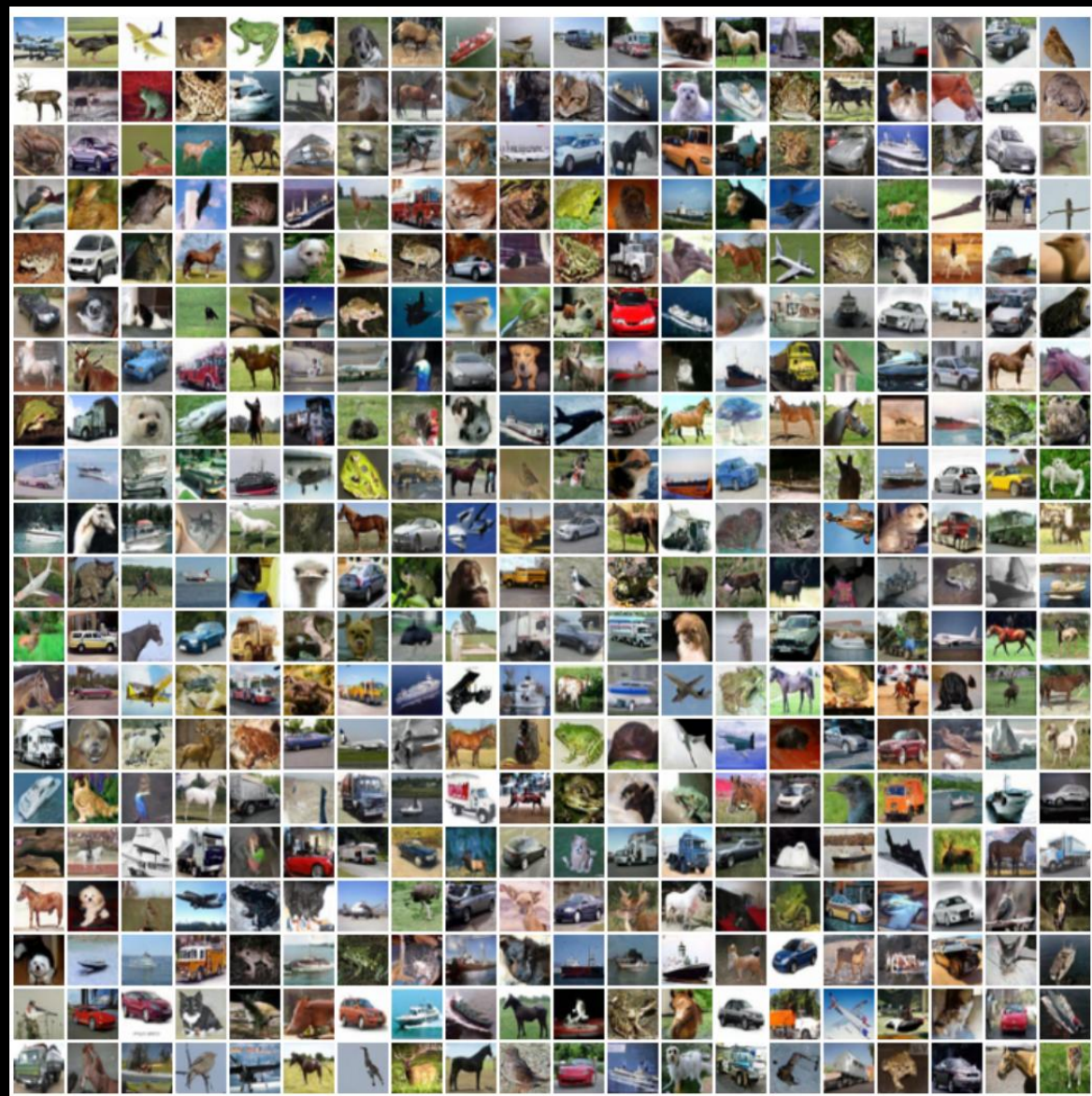
1. extreme flexibility in model structure,
2. exact sampling,

¹Non-parametric methods can be seen as transitioning smoothly between tractable and flexible models. For instance, a non-parametric Gaussian mixture model will represent a small amount of data using a single Gaussian, but may represent infinite data as a mixture of an infinite number of Gaussians.

Sample Generation Processes

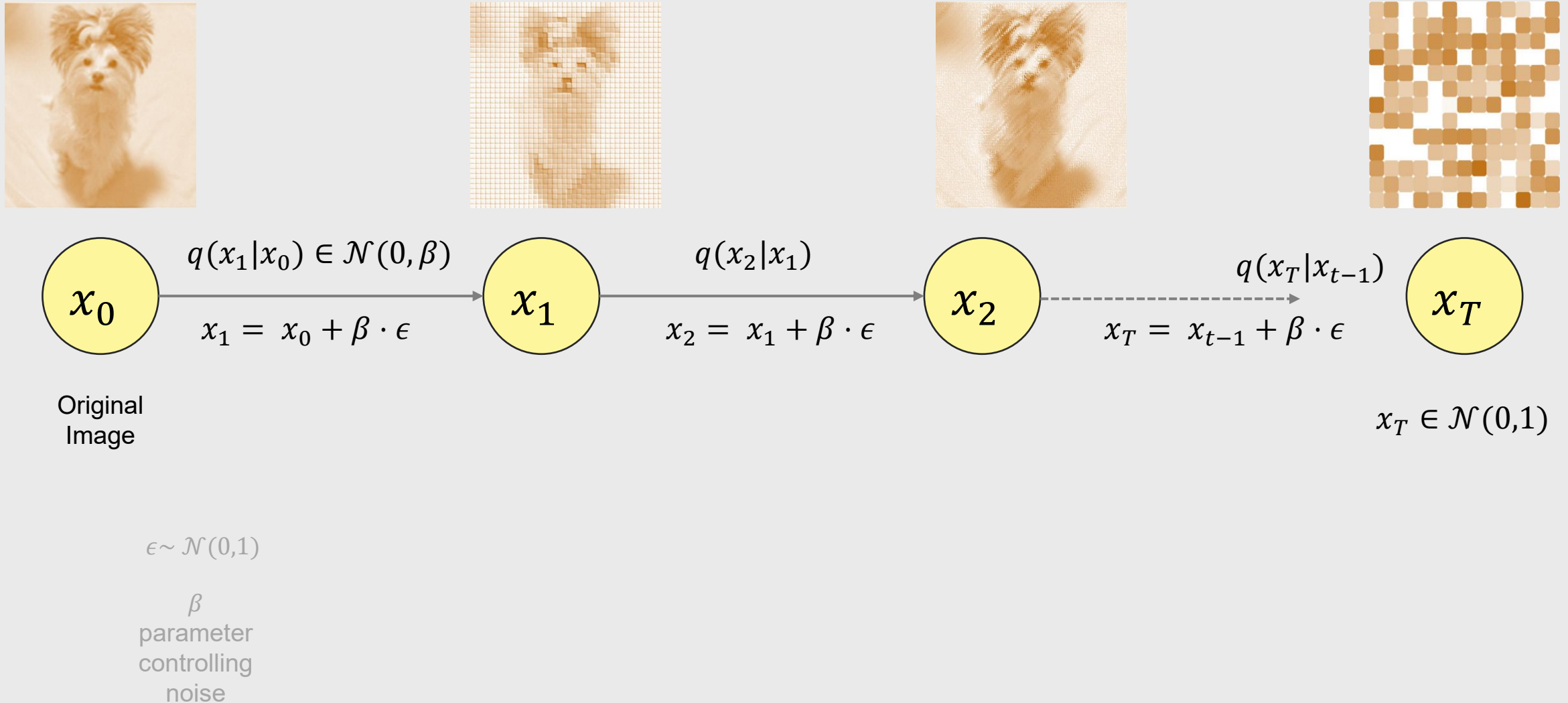


Generated Samples

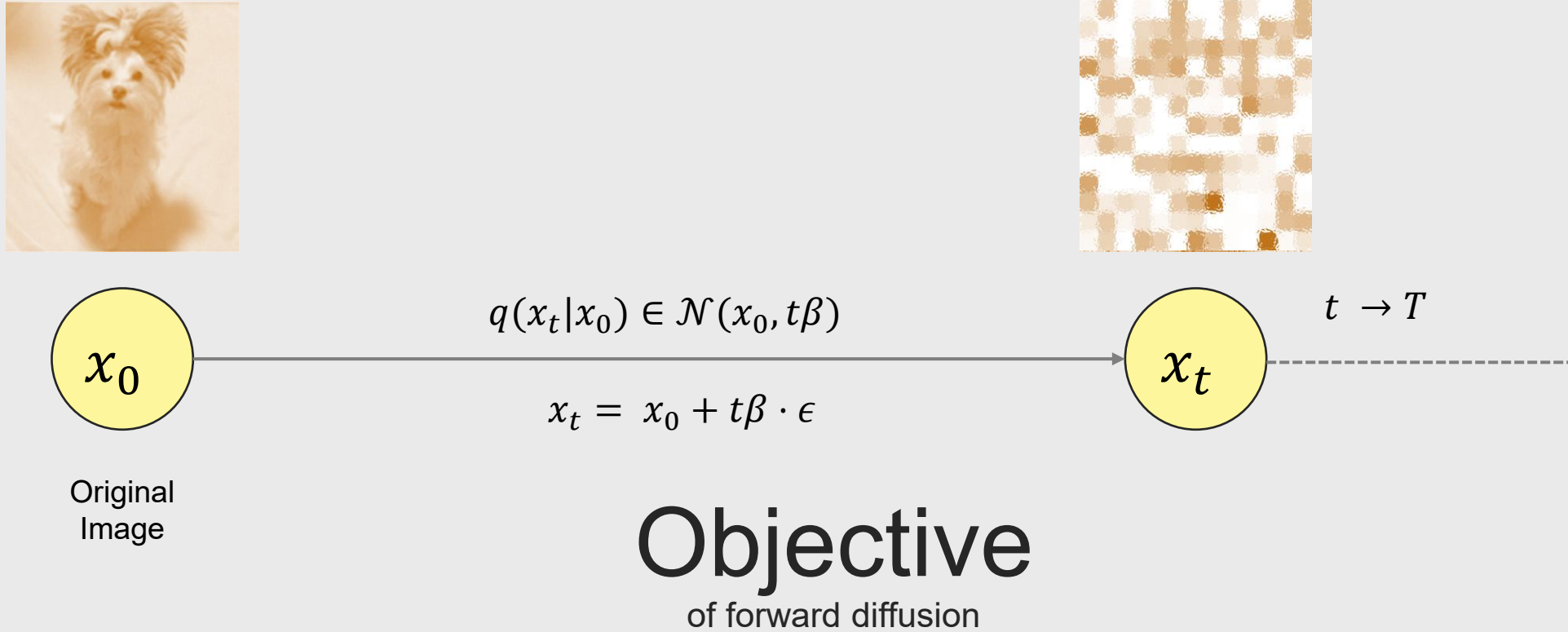


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Loss function in diffusion model

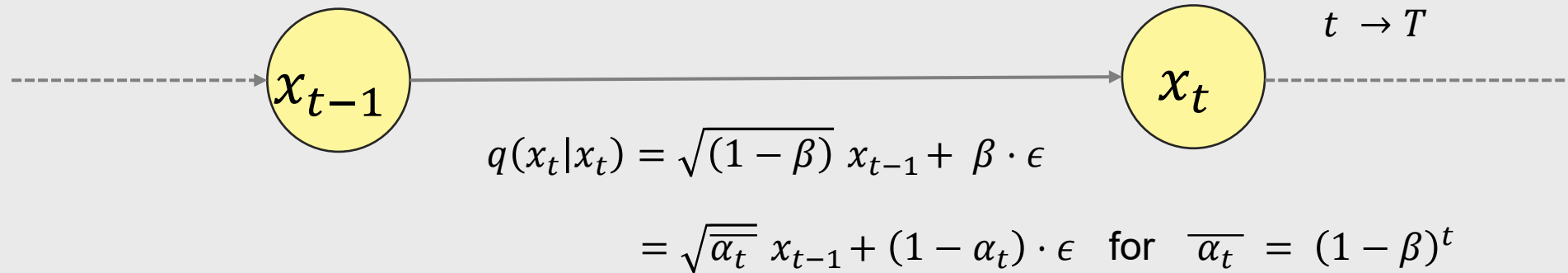
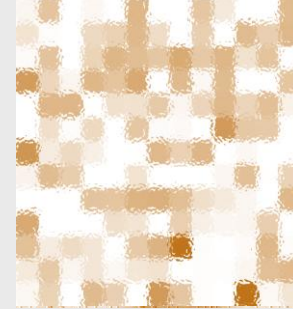


Loss function in diffusion model



$$q(x_t|x_0) \xrightarrow{t \rightarrow \infty} \mathcal{N}(0,1)$$

Loss function in diffusion model

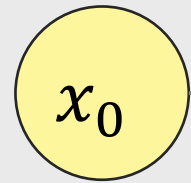
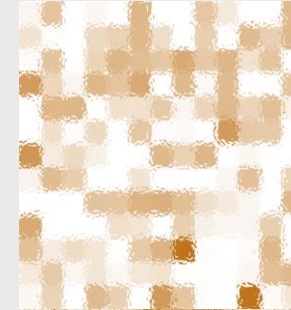


Loss function in diffusion model

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Objective of forward diffusion

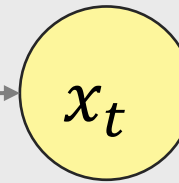


Original
Image

$$q(x_t|x_0) \xrightarrow{t \rightarrow \infty} \mathcal{N}(0,1)$$

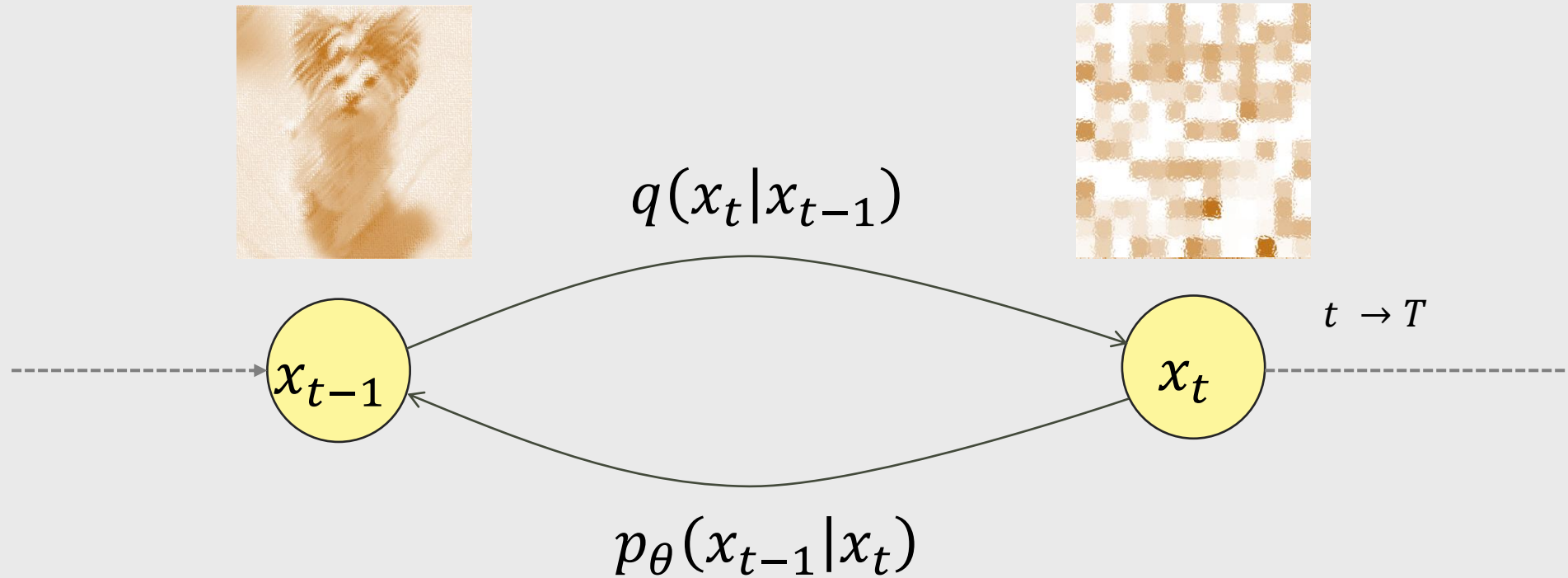
$$q(x_t|x_0) = \sqrt{(1-\beta)} x_0 + \beta \cdot \epsilon$$

$$= \sqrt{\bar{\alpha}_t} x_0 + (1 - \alpha_t) \cdot \epsilon \quad \text{for} \quad \bar{\alpha}_t = (1 - \beta_t)^t = \prod_{i=1}^t (1 - \beta_i)$$



$t \rightarrow T$

Loss function in diffusion model



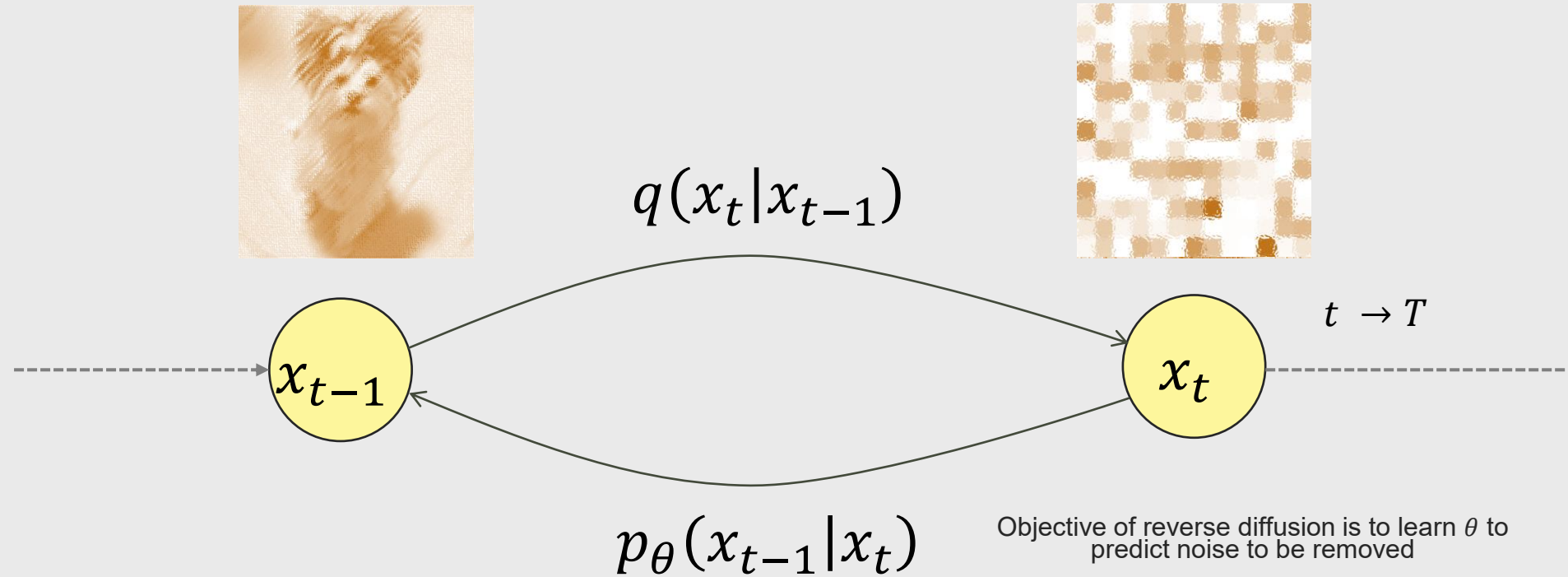
Objective

of reverse diffusion – learn θ to predict noise of previous step and

Minimise

$$-\log p_\theta(x_0)$$

Loss function in diffusion model



Complete forward diffusion processes

$$q(x_t \dots x_1 | x_0) = q(x_1 | x_0) q(x_2 | x_1) \dots q(x_T | x_{t-1}) = \prod_t^T q(x_t | x_{t-1})$$

Loss function in diffusion model

$$q(x_{1:T} | x_0) = \prod_t^T q(x_t | x_{t-1})$$

Complete forward diffusion processes

$$p_\theta(x_{0:T}) = p_\theta(x_T) \prod_t^T q(x_{t-1} | x_t)$$

Complete reverse diffusion processes

Objective of reverse diffusion – learn θ to predict noise of previous step to **minimise**

$$-\log p_\theta(x_0) = -\log \int p_\theta(x_{0:T}) dx_{1:T}$$

$$-\log p_\theta(x_0) = -\log \int q(x_{1:T} | x_0) \frac{p_\theta(x_{0:T})}{q(x_{1:T} | x_0)} dx_{1:T}$$

$$-\log p_\theta(x_0) \leq -\mathbb{E}_{q(x_{1:T} | x_0)} \left[\log \frac{p_\theta(x_{0:T})}{q(x_{1:T} | x_0)} \right]$$

Loss function in diffusion model

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$$L \leq -\mathbb{E}_{q(x_{1:T} | x_0)} \left[\log \frac{p_\theta(x_{0:T})}{q(x_{1:T} | x_0)} \right]$$

$$L = -\mathbb{E}_{q(x_{1:T} | x_0)} \left[\sum_{t>1} D_{\text{KL}}(q(x_{t-1} | x_t, x_0) || p_\theta(x_{0:T})) \right]$$

$$L = -\mathbb{E}_{q(x_{1:T} | x_0)} \left[\sum_{t>1} \frac{1}{2\sigma_t^2} \|\mu_t - \mu_\theta(x_t, t)\|^2 \right]$$

$$\mu_t = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{(1 - \bar{\alpha}_t)} \epsilon \right) \qquad \mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{(1 - \bar{\alpha}_t)} \epsilon_\theta(x_t, t) \right)$$

$$L = -\mathbb{E}_{q(x_{1:T} | x_0)} \left[\sum_{t>1} \frac{\beta_t^2}{2\sigma_t^2 \alpha_t (1 - \bar{\alpha}_t)} \|\epsilon - \epsilon_\theta(x_t, t)\|^2 \right]$$

Generative AI

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